

Fractional SPDE with Time-independent Noise: Solvability and Exact Asymptotics*

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Abstract

In this article, we study the

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1 Introduction

In this paper we study the following fractional stochastic equation:

$$\begin{cases} (\partial_t^b + \frac{\nu}{2}(-\Delta)^{a/2}) u(t, x) = I_t^r \left[\sqrt{\theta} u(t, x) \dot{W}(x) \right] & x \in \mathbb{R}^d, t > 0 \\ u(0, \cdot) = 1 & b \in (0, 1] \\ u(0, \cdot) = 1, \quad \partial_t u(0, \cdot) = 0 & b \in (1, 2) \end{cases} \quad (1.1) \quad \boxed{\text{E:SPDE}}$$

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where $a \in (0, 2)$, $b \in (0, 2)$, $r \geq 0$, $\nu > 0$ and $\theta > 0$. Here the noise $W = \{W(\varphi) : \varphi \in \mathcal{D}(\mathbb{R}^d)\}$ is a centered and time-independent Gaussian process, defined on a complete probability space $(\Omega, \mathcal{F}, P)$, with mean zero and covariance

$$\mathbb{E}[W(\varphi)W(\psi)] = \int_{\mathbb{R}^d} \mathcal{F}\varphi(\xi)\overline{\mathcal{F}\psi(\xi)}\mu(d\xi) =: \langle \varphi, \psi \rangle_{\mathcal{H}},$$

where μ refers to the *spectral measure*, which is assumed to be a nonnegative and nonnegative definite tempered measure on $\mathbb{R}^d$. Let γ be the Fourier transform of μ , which is also a nonnegative and nonnegative measure on $\mathbb{R}^d$ thanks to Bochner's theorem. Throughout of the paper, we use $\mathcal{F}\varphi(\xi) = \int_{\mathbb{R}^d} \exp(-ix\xi)\varphi(x)dx$ to denote the Fourier transform of a test function φ .

In (1.1), $(-\Delta)^{a/2}$ refers to the *fractional Laplacian* of order a , ∂_t^b denotes the *Caputo fractional differential operator*

$$\partial_t^b f(t) := \begin{cases} \frac{1}{\Gamma(m-b)} \int_0^t d\tau \frac{f^m(\tau)}{(t-\tau)^{b+1-m}} & \text{if } m-1 < b < m, \\ \frac{d^m}{dt^m} f(t) & \text{if } b = m, \end{cases}$$

where m is an integer, and I_t^r refers to the *Riemann-Liouville fractional integral* of order $r > 0$

$$I_t^r f(t) := \frac{1}{\Gamma(r)} \int_0^t (t-s)^{r-1} f(s)ds, \quad \text{for } t > 0,$$

with the convention that when $r = 0$, $I_t^0 = \text{Id}$ reduces to the identity operator. The fundamental solutions to (1.1) consist of a triplet:

$$\left\{ Z_{a,b,r,\nu,d}(t, x), Z_{a,b,r,\nu,d}^*(t, x), Y_{a,b,r,\nu,d}(t, x) : t > 0, x \in \mathbb{R}^d \right\}.$$

Each member of the triplet can be expressed explicitly in terms of the *Fox H-function* which is much more complicated than the fundamental function for either the heat or wave equation; See Theorem 4.1 of [5]. Since we are interested in the constant one initial condition (and zero initial velocity when $b > 1$), Theorem 4.1 (*ibid.*) implies that the corresponding solution to the homogeneous equation (i.e. the solution when there is no driving source) is equal to the constant one. Hence, by setting $G(t, x) := Y_{a,b,r,\nu,d}(t, x)$, (1.1) can be written as the following integral equation:

$$u(t, x) = 1 + \sqrt{\theta} \int_0^t \left(\int_{\mathbb{R}^d} G(t-s, x-y) u(s, y) W(\delta y) \right) ds, \quad (1.2)$$

E:spdeint

where the stochastic integral is in the *Skorohod* sense; see Definition 2.1 below.

Regarding the noise, we aim to cover the Riesz kernel case, the fractional noise and a mixture of them, which is stated in the following assumption:

A:Noise

Assumption 1.1. Let $k \in \{1, \dots, d\}$ and partition the d -coordinates of $x = (x_1, \dots, x_d)$ into k distinct groups of size d_i so that $d_1 + \dots + d_k = d$. Denote $x_{(i)} = (x_{i_1}, \dots, x_{i_{d_i}})$ to be the coordinates in the i^{th} partition. Assume that the correlation function of the Gaussian noise is given by

$$\gamma(x) = \prod_{i=1}^k |x_{(i)}|^{-\alpha_i} \quad (1.3)$$

E:corfuncti

where $\alpha_i \in (0, d_i)$. Define $\alpha := \sum_{i=1}^k \alpha_i$.

The spectral density corresponding to (1.3) is given by $\mu(d\xi) = \varphi(\xi)d\xi$ with

$$\varphi(\xi) = \prod_{i=1}^k C_{\alpha_i, d_i} |\xi_{(i)}|^{-(d_i - \alpha_i)}. \quad (1.4) \quad \boxed{\text{E:specfunct.}}$$

Moreover, one can find a nonnegative and nonnegative definite K such that $\gamma = K * K$ where ‘ $*$ ’ denotes the spatial convolution. Indeed, one can choose

$$K(x) = \prod_{i=1}^k \beta_{\alpha_i, d_i} |x_i|^{-(d_i - \alpha_i)}. \quad (1.5) \quad \boxed{\text{E:K}}$$

The two constants in both (1.4) and (1.5) are defined as

$$C_{\alpha, d} = \pi^{-d/2} 2^{-\alpha} \frac{\Gamma((d - \alpha)/2)}{\Gamma(\alpha/2)} \quad \text{and} \quad \beta_{\alpha, d} = \pi^{-d/4} \frac{\Gamma((d + \alpha)/4)}{\Gamma((d - \alpha)/4)} \sqrt{\frac{\Gamma((d - \alpha)/2)}{\Gamma(\alpha/2)}}. \quad (1.6) \quad \boxed{\text{E:noisecons}}$$

Example 1.2. The case $k = 1$ in (1.3) (or (1.4)) recovers the Riesz kernel case. In this case,

$$\gamma(x) = |x|^{-\alpha}, \quad \varphi(x) = C_{\alpha, d} |x|^{-(d - \alpha)} \quad \text{and} \quad K(x) = \beta_{\alpha, d} |x|^{-(d + \alpha)/2}.$$

Example 1.3. The case when $k = d$ in (1.3) (or (1.4)) recovers the *time-independent fractional noise*. In this case,

$$\gamma(x) = \prod_{i=1}^d |x_i|^{-\alpha_i}, \quad \varphi(\xi) = \prod_{i=1}^d C_{\alpha_i, 1} |\xi_i|^{-(1 - \alpha_i)} \quad \text{and} \quad K(x) = \prod_{i=1}^d \beta_{\alpha_i, 1} |x_i|^{-(1 + \alpha_i)/2}.$$

The stochastic heat equation with such noise was earlier studied by Hu [9].

Next, we need to make some assumption on the parameters a, b, r of (1.1) as well as the dimension d . Under these assumptions, the fundamental function G will be nonnegative and smooth for $x \neq 0$; see Theorems 4.1 and 4.6 of [4].

Nonnegative

Assumption 1.4. The parameters a, b, r and the dimension d must satisfy one of the following conditions:

1. $d \geq 1$, $b \in (0, 1)$, $a \in (0, 2]$, $r \geq 0$;
2. $d \geq 1$, $b = 1$, $a \in (0, 2]$, $r = 0$ or $r > 1$;
3. $1 \leq d \leq 3$, $1 < b < a \leq 2$, $r > 0$;
4. $1 \leq d \leq 3$, $1 < b = a < 2$, $r > \frac{d + 3}{2} - b$.

We now introduce the following variational constant, which plays an important role in this paper:

$$\mathcal{M}_a(\Theta f, \theta) := \sup_{g \in \mathcal{F}_a} \left\{ \langle g^2 * g^2, f \rangle^{1/2} - \frac{\theta}{2} \mathcal{E}_a(g, g) \right\} \quad \text{and} \quad \mathcal{M}_a := \mathcal{M}_a(| \cdot |^{-\alpha}, 1);$$

see Section 2.2 for more notation and details. Our goal is to develop exact moment asymptotics to the solution $u(t, x)$ so that it can interpolate the known results for both stochastic heat and wave equations; see Remark 1.7 below. We follow the techniques laid out in [1] where the stochastic wave equation, namely, $a = b = \nu = 2$ and $r = 0$, has been exclusively studied. This paper acts as a generalization of that paper to the fraction stochastic equation. Here is the main result of this paper.

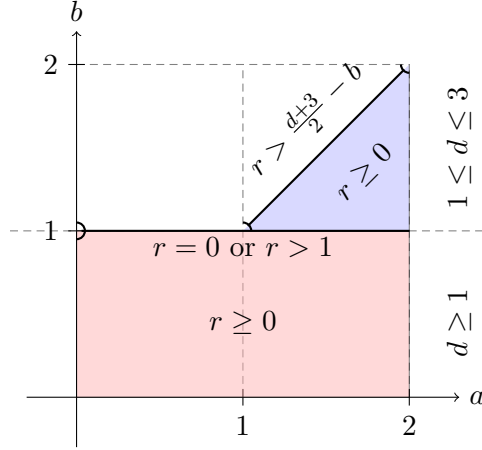


Figure 1: Illustration of the sufficient conditions (Assumption 1.4) for $G(t, \cdot)$ to be nonnegative.

Theorem 1.5. Under Assumption 1.4, equation (1.1) driven by the noise with spectral density given in equation (1.4) has a unique solution $u(t, x)$ in $L^p(\Omega)$ for all $p \geq 2$, $t > 0$, and $x \in \mathbb{R}^d$ provided that

$$\alpha < \min \left(\frac{a}{b} [2(b+r) - 1], 2a, d \right). \quad (1.7)$$

Theorem 1.6. Under the setting of Theorem 1.5, we have the following moment asymptotics:

$$\begin{aligned} \lim_{t_p \rightarrow \infty} t_p^{-\beta} \log \|u(t, x)\|_p &= \left(\frac{1}{2} \right) \left(\frac{2a}{2a(b+r) - b\alpha} \right)^\beta \\ &\times \left(\theta \nu^{-\alpha/a} \mathcal{M}_a^{\frac{2a-\alpha}{a}} \right)^{\frac{a}{2a(b+r) - b\alpha - a}} \left(2(b+r) - \frac{b\alpha}{a} - 1 \right), \end{aligned} \quad (1.8)$$

where

$$\beta := \frac{2(b+r) - \frac{b\alpha}{a}}{2(b+r) - \frac{b\alpha}{a} - 1} \quad \text{and} \quad t_p^\beta := (p-1)^{\beta-1} t^\beta. \quad (1.9)$$

In particular, for any $p \geq 2$ and $t > 0$ fixed, we have

$$\begin{aligned} \lim_{t \rightarrow \infty} t^{-\beta} \log \mathbb{E}(|u(t, x)|^p) &= p(p-1)^{\frac{1}{2(b+r) - \frac{b\alpha}{a} - 1}} \left(\frac{1}{2} \right) \left(\frac{2a}{2a(b+r) - b\alpha} \right)^\beta \\ &\times \left(\theta \nu^{-\alpha/a} \mathcal{M}_a^{\frac{2a-\alpha}{a}} \right)^{\frac{a}{2a(b+r) - b\alpha - a}} \left(2(b+r) - \frac{b\alpha}{a} - 1 \right), \end{aligned} \quad (1.10)$$

and

$$\begin{aligned} \lim_{p \rightarrow \infty} p^{-\beta} \log \mathbb{E}(|u(t, x)|^p) &= t^\beta \left(\frac{1}{2} \right) \left(\frac{2a}{2a(b+r) - b\alpha} \right)^\beta \\ &\times \left(\theta \nu^{-\alpha/a} \mathcal{M}_a^{\frac{2a-\alpha}{a}} \right)^{\frac{a}{2a(b+r) - b\alpha - a}} \left(2(b+r) - \frac{b\alpha}{a} - 1 \right), \end{aligned} \quad (1.11)$$

respectively.

Proof. The asymptotics in (1.10) and (1.11) are immediate consequences of (1.8). The matching upper and lower bounds for the limit in (1.8) are proved in Theorems 4.1 and 5.1 below, respectively. $\square$

R:Inter

Remark 1.7. The asymptotics in (1.8) interpolates the corresponding results for the stochastic wave and heat equations:

(1) By setting

$$a = b = \nu = 2 \quad \text{and} \quad r = 0,$$

we have that $t_p = (p-1)^{1/(4-\alpha)}t$, and results in (1.8), (1.10), and (1.11) recover the corresponding results for the stochastic wave equation, namely, (1.9), (1.10), and (1.11) of [1], respectively.

(2) As for the stochastic heat equation case, we are not aware of the exact asymptotics results for SHE with time-independent noise case. However, as explained in Remark 1.9 of [1], one may apply Chen's asymptotic results in [6] for SHE with time-dependent noise, by setting $H_0 = 1$, to obtain the corresponding time-independent results. In particular, by setting

$$a = b = 2, \quad \nu = 1 \quad \text{and} \quad r = 0,$$

we have that $t_p = (p-1)^{2/(4-\alpha)}t$, and results in (1.8) and (1.10) recover the corresponding results for the stochastic heat equation, namely, (1.16) and (1.17) of [1], respectively.

In the following, we first give some preliminaries in Section 2 including the Malliavin calculus, the definition of the solution, and some variational constants. The existence and uniqueness of a $L^p(\Omega)$ solution — Theorem 1.5 — is proven in Section 3. The upper bound and lower bound for (1.8) are established in Sections 4 and 5, respectively. Finally, in the appendix — Section 6, we list a few proofs of the results in the paper.

2 Some preliminaries

S:Pre

2.1 Malliavin Calculus

S:Malliavin

To briefly summarize [8], we start with a nonnegative and nonnegative definite tempered measure Γ with density γ in the sense that $\Gamma(dx) = \gamma(x)dx$, and

$$\int_{\mathbb{R}^d} \Gamma(dx) (\varphi * \tilde{\varphi})(x) \geq 0 \quad \text{for all } \varphi \in \mathcal{S}(\mathbb{R}^d)$$

where $\tilde{\varphi}(x) = \varphi(-x)$. In addition we require the existence of an $r > 0$ such that

$$\int_{\mathbb{R}^d} \Gamma(dx) \frac{1}{(1 + |x|^2)^r} < \infty.$$

According to the Bochner theorem, there exists a nonnegative measure μ , often referred to as the spectral measure on $\mathbb{R}^d$ whose Fourier transform is Γ in the sense that for any $\varphi \in (\mathbb{R}^d)$

$$\int_{\mathbb{R}^d} \Gamma(dx) \varphi(x) = \int_{\mathbb{R}^d} \mu(dt) \mathcal{F}\varphi(t)$$

The functional defined on the space of test functions $\mathcal{D}(\mathbb{R}^d)$

$$C(\varphi, \psi) = \int_{\mathbb{R}} \mathcal{F}\varphi(\xi) \overline{\mathcal{F}\psi(\xi)} \mu(d\xi)$$

is nonnegative define and thus we may associate it as the covariance functional of a zero-mean Gaussian processes, $W := \{W(\varphi) : \varphi \in \mathcal{D}(\mathbb{R}^d)\}$. In other words,

$$\mathbb{E}(W(\varphi)W(\psi)) = \int_{\mathbb{R}} \mathcal{F}\varphi(\xi) \overline{\mathcal{F}\psi(\xi)} \mu(d\xi) =: \langle \varphi, \psi \rangle_{\mathcal{H}}.$$

Let $\mathcal{H}$ be the completion of $\mathcal{D}(\mathbb{R}^d)$ with respect to $\langle \cdot, \cdot \rangle_{\mathcal{H}}$ and thus we see $\varphi \mapsto W(\varphi)$ is an isometry from the normed space $(\mathcal{D}(\mathbb{R}^d), \|\cdot\|_{\mathcal{H}})$ to $L^2(\Omega)$ where $\|f\|_{\mathcal{H}} = \langle f, f \rangle_{\mathcal{H}}$.

We denote δ the Skorohod integral with respect to W and we denote its domain $\text{Dom}(\delta)$. For any $u \in \text{Dom}(\delta)$, we write

$$\delta(u) = \int_{\mathbb{R}^d} u(x) W(dx).$$

Through construction of δ , $\delta(u)$ is a random variable in $L^2(\Omega)$. For a deeper explanation, we direct the reader to Nualart [11].

D:Sol

Definition 2.1. A random field $u = \{u(t, x) : t \geq 0, x \in \mathbb{R}^d\}$ such that $\|u(t, x)\|_2 < \infty$ for all $t \geq 0$ and $x \in \mathbb{R}^d$ is called a *mild solution* to the equation (1.1) if the random field $\{G(t-s, x-y)u(s, y) : s \in [0, t], y \in \mathbb{R}^d\}$ is Skorohod integrable for each $s \in [0, t]$ and $x \in \mathbb{R}^d$ fixed and satisfies the following integral equation:

$$u(t, x) = 1 + \sqrt{\theta} \int_0^t \left(\int_{\mathbb{R}^d} G(t-s, x-y) u(s, y) W(dy) \right) ds. \quad (2.1)$$

Through the standard Picard iteration scheme, the solution can be expressed by

$$u(t, x) = 1 + \sum_{k=1}^{\infty} \theta^{k/2} I_k(f_k(\cdot, x, t)) \quad (2.2)$$

E:solwexp

where $I_k : \mathcal{H}^{\otimes k}((\mathbb{R}^d)^k) \rightarrow \mathcal{H}_k$ is the k^{th} order Skorohod integral and $\mathcal{H}_k$ is the k^{th} Wiener chaos space. The kernels $f_k(\cdot, x, t) \in \mathcal{H}^{\otimes k}((\mathbb{R}^d)^k)$ are calculated through the Piccard iteration scheme to be

$$\begin{aligned} f_n(x_1, \dots, x_k; x, t) &= \int_0^t \int_0^{t_n} \cdots \int_0^{t_2} G(t-t_n, x-x_n) \cdots G(t_2-t_1, x_2-x_1) dt_1 \cdots dt_n \\ &= \int_0^t \int_0^{t_n} \cdots \int_0^{t_2} G(t_1, x-x_n) \cdots G(t_n-t_{n-1}, x_2-x_1) dt_1 \cdots dt_n. \end{aligned} \quad (2.3)$$

E:wker

For ease of notation throughout this article, we write the above integrals as

$$\begin{aligned} f_n(x_1, \dots, x_n; x, t) &= \int_{[0, t]_{<}^n} G(t-t_n, x-x_n) \cdots G(t_2-t_1, x_2-x_1) d\vec{t} \\ &= \int_{[0, t]_{<}^n} G(t_1, x-x_n) \cdots G(t_n-t_{n-1}, x_2-x_1) d\vec{t}. \end{aligned}$$

where in general $d\vec{z} = \prod_{k=1}^d dz_k$ and the dimension will always be easily understood. We denote by $\widetilde{f}_n(\cdot, x, t)$ to be the symmetrization of $f_n(\cdot, x, t)$ in the variables $x_1, \dots, x_n$:

$$\widetilde{f}_n(\cdot, x, t) = \frac{1}{n!} \sum_{\rho \in \Sigma_n} f_n(x_{\rho(1)}, \dots, x_{\rho(n)}) \quad (2.4)$$

$$= \frac{1}{n!} \sum_{\rho \in \Sigma_n} \int_{[0,t]_{<}^n} G(t - t_n, x - x_{\rho(n)}) \cdots G(t_2 - t_1, x_{\rho(2)} - x_{\rho(1)}) d\vec{t} \quad (2.5)$$

where Σ_n is the set of all permutations of $\{1, \dots, n\}$. In particular, for $x = 0$ we have

$$\widetilde{f}_n(\cdot; 0, t) = \frac{1}{n!} \sum_{\rho \in \Sigma_n} \int_{[0,t]_{<}^n} G(t_1, x_{\rho(1)}) \cdots G(t_n - t_{n-1}, x_{\rho(n)} - x_{\rho(n-1)}) d\vec{t}. \quad (2.6) \quad \text{E:fperm0}$$

By setting $t_{n+1} = t$, the Fourier transform of the kernels, f_n , is given by

$$\mathcal{F}f_n(\cdot; x, t)(\xi_1, \dots, \xi_n) = e^{-i(\sum_{j=1}^n \xi_j) \cdot x} \int_{[0,t]_{<}^n} \prod_{k=1}^n \overline{\mathcal{F}G(t_{k+1} - t_k, \cdot)} \left(\sum_{j=1}^k \xi_j \right) dt. \quad (2.7) \quad \text{E:fweker}$$

Recall the notation above that $G(t, x) = Y_{a,b,r,d}(t, x)$. The following result shows an important scaling property that both $\mathcal{F}G(t, \cdot)(\xi)$ and $\left\| \widetilde{f}_n(\cdot, t, x) \right\|_{\mathcal{H}^{\otimes n}}$ have.

L:Scale **Lemma 2.2.** *For any $c > 0$ and $t > 0$, the two following scaling properties hold:*

$$\mathcal{F}G(t, \cdot)(c\xi) = c^{-\frac{a}{b}(b+r-1)} \mathcal{F}G\left(c^{\frac{a}{b}}t, \cdot\right)(\xi) \quad \text{and} \quad \mathcal{F}G(ct, \cdot)(\xi) = c^{b+r-1} \mathcal{F}G(t, \cdot)(c^{b/a}\xi), \quad (2.8) \quad \text{E:green-scal}$$

and

$$\left\| \widetilde{f}_n(\cdot, 0, t) \right\|_{\mathcal{H}^{\otimes n}}^2 = t^{[2(b+r)-b\alpha/a]n} \left\| \widetilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2. \quad (2.9) \quad \text{E:normscale}$$

Proof. The scaling properties in (2.8) is a direct consequence of the explicit expression of $\mathcal{F}G(t, \cdot)(\xi)$ as in (4.8) of [5]. Property (2.9) is a direct consequence of (2.8) and the scaling property of the spectral measure μ . We leave the details for the interested readers. $\square$

Recall the following standard result about the existence and uniqueness of the solution when the solution can be written as the Wiener chaos expansion as in (2.2).

Theorem 2.3. *Fix any $T \in (0, \infty]$. Suppose that $f_n(\cdot, x, t) \in \mathcal{H}^{\otimes n}$ for any $t \in (0, T)$, $x \in \mathbb{R}^d$ and $n \geq 1$. Then equation (1.1) has a unique $L^2(\Omega)$ solution on $(0, T) \times \mathbb{R}^d$ if and only if the series (2.2) converges in $L^2(\Omega)$ for any $(t, x) \in (0, T) \times \mathbb{R}^d$, which is equivalent to the convergence of the series (2.10). In this case, the solution is given by (2.2) with the second moment given by*

$$\mathbb{E}(u(t, x)^2) = \sum_{n \geq 0} \theta^n n! \left\| \widetilde{f}_n(\cdot, x, t) \right\|_{\mathcal{H}^{\otimes n}}^2 \quad \text{for all } (t, x) \in (0, T) \times \mathbb{R}^d. \quad (2.10) \quad \text{E:2mom}$$

2.2 Some large deviation results

SS:LD

Let us first define

$$\rho := \rho_{\nu, a} = \sup_{\|f\|_2=1} \int_{\mathbb{R}^d} \left[\int_{\mathbb{R}^d} \frac{f(x+y)f(y)}{\sqrt{1 + \frac{\nu}{2}|x+y|^a} \sqrt{1 + \frac{\nu}{2}|y|^a}} dy \right]^2 \mu(dx).$$

Recall that when $\mu(d\xi) = C_{\alpha,d} |\xi|^{-(d-\alpha)} d\xi$ which corresponds to the Riesz kernel $\gamma(x) = |x|^{-\alpha}$ for some $\alpha \in (0, d)$, Bass, Chen and Rosen [3] established that when $a \in (0, 2]$ and $\nu = 2$,

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \left[\frac{1}{(n!)^2} \int_{(\mathbb{R}^d)^n} \left(\sum_{\sigma \in \Sigma_n} \prod_{k=1}^n \frac{1}{1 + \left| \sum_{j=k}^n \xi_{\sigma(j)} \right|^a} \right)^2 \mu(d\vec{\xi}) \right] = \log(\rho_{2,a}), \quad (2.11) \quad \text{E:rhowavede}$$

and

$$\rho_{2,a} = \Lambda_\alpha^{2-(\alpha/a)} \quad (2.12) \quad \text{E:oldrhovar}$$

where

$$\Lambda_\alpha := \sup_{g \in \mathcal{F}_a} \left\{ \left(\int_{\mathbb{R}^{2d}} \frac{g^2(x_1)g^2(x_2)}{|x_1 + x_2|^\alpha} d\vec{x} \right)^{1/2} - \mathcal{E}_a(g, g) \right\} \quad (2.13)$$

$$= \sup_{g \in \mathcal{F}_a} \left\{ \langle g^2 * g^2, |\cdot|^{-\alpha} \rangle_{L^2(\mathbb{R}^d)} - \mathcal{E}_a(g, g) \right\} \quad \text{and}$$

$$\mathcal{E}_a(g, g) := (2\pi)^{-d} \int_{\mathbb{R}^d} |\xi|^a |\mathcal{F}g(\xi)|^2 d\xi. \quad (2.14)$$

The following result shows the relationship between $\rho_{2,a}$ and ρ and well as the generalization of (2.11) for ρ .

L:rhovar

Lemma 2.4. *For the Riesz kernel case, i.e., when $\gamma(x) = |x|^{-\alpha}$ for some $\alpha \in (0, d)$, it holds that*

$$\rho_{\nu,a} = \left(\frac{\nu}{2}\right)^{-\alpha/a} \Lambda_\alpha^{2-(\alpha/a)}, \quad \text{for any } \nu > 0 \text{ and } a \in (0, 2]. \quad (2.15) \quad \text{E:rhoLam}$$

In addition,

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \left[\frac{1}{(n!)^2} \int_{(\mathbb{R}^d)^n} \left(\sum_{\sigma \in \Sigma_n} \prod_{k=1}^n \frac{1}{1 + \frac{\nu}{2} |\sum_{j=k}^n \xi_{\sigma(j)}|^a} \right)^2 \mu(d\vec{\xi}) \right] = \log(\rho_{\nu,a}). \quad (2.16) \quad \text{E:LogRho}$$

This lemma can be proved in the standard scaling arguments together with (2.12). For completeness, the proof is presented in the Appendix of the paper.

Next we show that ρ is finite. We do this by showing $\rho_{2,a}$ is finite and then noting that this implies that ρ is finite by 2.4. We stated in the introduction that Bass, Chen and Rosen proved in [3] that ρ is finite for the case of the Riesz kernel. We mimic his proof but were he uses the Hardy-Littlewood-Sobolev theorem, we make use of the weak Young's inequality instead.

L:rhofinite

Lemma 2.5. *For any f, g, h with $h \geq 0$ and for φ given as in equation (1.4), we have the following:*

$$\left(\int_{\mathbb{R}^d} \left[\int_{\mathbb{R}^d} \frac{|f(x+y)g(y)|}{\sqrt{h(x+y)}\sqrt{h(y)}} dy \right]^p \varphi(x) dx \right)^{1/p} \leq C \|f\|_2 \|g\|_2 \|h^{-1}\|_{pd/\alpha}$$

where $\alpha = \sum_{i=1}^k \alpha_i$ with $k \in \{1, \dots, d\}$, $\alpha_i \in (0, d_i)$ and $\varphi(x) = \prod_{i=1}^k |x_{(i)}|^{-(d_i - \alpha_i)}$ where $x_{(i)} = (x_{i_1}, \dots, x_{i_{d_i}})$ is the i^{th} group of $x = (x_1, \dots, x_d)$.

Proof. By observing the proof of Lemma 1.6 [3], we notice that we only need to prove equation (1.27) which says

$$\int_{\mathbb{R}^d} \int_{\mathbb{R}^d} F(y)G(x)\varphi(x-y)dydx \leq C \|F\|_{2d/(d+\alpha)} \|G\|_{2d/(d+\alpha)}. \quad (2.17) \quad \text{E:fracrhofin}$$

where

$$F(x) = \frac{|f(x)|}{(h(x))^{p/2}} \quad \text{and} \quad G(x) = \frac{|g(x)|}{(h(x))^{p/2}}$$

If we can do this then the result follows by copying the rest of Chen's proof. To prove (2.17), we use the weak version of Young's inequality. Recall that weak Young's inequality [10] says

$$\left| \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} a(x)b(x-y)c(y)dx dy \right| \leq K_{p,q,r,d} \|a\|_p \|b\|_{q,w} \|c\|_r$$

where $p, q, r > 1$ and $1/p + 1/q + 1/r = 2$ and

$$\|b\|_{q,w} = \sup_A |A|^{-1/q'} \int_A |f(x)| dx$$

where A is an arbitrary Borel set of finite measure. So to prove (2.17) we wish to be able to apply this with

$$p = r = 2d/(d + \alpha) \quad \text{and} \quad q = d / \left(d - \sum_{j=1}^d \alpha_j \right) = d/(d - \alpha)$$

and, in addition, with

$$a = F, \quad c = G \quad \text{and} \quad b = \varphi$$

Again, by following the steps of the proof from the Xia Chen paper, we note that it suffices to show that $\|\varphi\|_{q,w}$ is finite with $q = d/(d - \alpha)$. Note that $1/q' = 1 - 1/q = \alpha/d$.

First we define $A_R := A_1 \times \cdots \times A_k$ where $A_i \subset \mathbb{R}^{d_i}$ and $A_i = [-R, R]$ if $d_i = 1$ and $A_i = B_R(0)$ if $d_i > 1$ where $B_R(0)$ is the ball centered at the origin with radius R . With this we have that

$$\int_{A_i} |x_{(i)}|^{-(d_i - \alpha_i)} dx_{(i)} = \begin{cases} 2 \frac{R^{\alpha_i}}{\alpha_i} & , d_i = 1 \\ |\mathbb{S}^{d_i-1}| \frac{R^{\alpha_i}}{\alpha_i} & , d_i > 1 \end{cases} \quad (2.18) \quad \boxed{\text{E:int}}$$

where we used polar coordinated to calculate the integral when $d_i > 1$ and $|\mathbb{S}^{d_i-1}|$ is the surface area of the sphere in $\mathbb{R}^{d_i}$. Also,

$$|A_i| = \begin{cases} 2R & , d_i = 1 \\ |\mathbb{S}^{d_i-1}| \frac{R^{d_i}}{d_i} & , d_i > 1. \end{cases} \quad (2.19) \quad \boxed{\text{E:leb}}$$

Recall that $1/q' = \alpha/d$. Then by combining (2.18) and (2.19) we see that

$$\begin{aligned} |A_R|^{-1/q'} \int_{A_R} \varphi(x) dx &= \left(\prod_{i=1}^k |A_i| \right)^{-1/q'} \prod_{i=1}^k \int_{A_i} |x_{(i)}|^{-(d_i - \alpha_i)} dx_{(i)} \\ &= R^{-\alpha} \left(\prod_{i=1}^k \begin{cases} 2 & , d_i = 1 \\ |\mathbb{S}^{d_i-1}|/d_i & , d_i > 1. \end{cases} \right) \times R^{\alpha} \left(\prod_{i=1}^k \begin{cases} 2/\alpha_i & , d_i = 1 \\ |\mathbb{S}^{d_i-1}|/\alpha_i & , d_i > 1 \end{cases} \right) \\ &= K \end{aligned}$$

where the constant K does not depend on R . Hence we that

$$\sup_{R>0} |A_R|^{-1/q'} \int_{A_R} \varphi(x) dx = K.$$

and by the symmetry of φ we can in fact say that

$$\sup_B |B|^{-1/q'} \int_B \varphi(x) dx = K$$

where B denotes the Borel sets of finite measure. Hence, $\varphi \in L_{q,w}(\mathbb{R}^d)$ with $q = d/(d - \alpha)$. This completes the proof of Lemma 2.5. $\square$

We finish this section by making a slight generalization of Λ_α . For this, we define

$$\mathcal{M}_a(\Theta f, \theta) = \sup_{g \in \mathcal{F}_a} \left\{ \langle g^2 * g^2, f \rangle^{1/2} - \frac{\theta}{2} \mathcal{E}_a(g, g) \right\}. \quad (2.20)$$

One notices that by making this generalization, we can apply Theorem 1.5 to the SPDE (1.1) driven by an arbitrary Gaussian noise with correlation function f as long as one can show that ρ is finite. By slightly modifying the proof of Lemma 2.3 [1], one can show that

$$\mathcal{M}_a(\Theta f, \theta) = \Theta^{\frac{a}{2a-\alpha}} \theta^{-\frac{\alpha}{2a-\alpha}} \mathcal{M}_a(f, 1). \quad (2.21)$$

Finally we note that

$$\Lambda_\alpha = \mathcal{M}_a(|\cdot|^{-\alpha}, 2) = 2^{-\frac{\alpha}{2a-\alpha}} \mathcal{M}_a(|\cdot|^{-\alpha}, 1) \quad (2.22)$$

and by defining $\mathcal{M}_a := \mathcal{M}_a(|\cdot|^{-\alpha}, 1)$ and by applying Theorem 2.4 we see that

$$\rho = \left(\frac{\nu}{2}\right)^{-\alpha/a} \left[2^{-\frac{\alpha}{2a-\alpha}} \mathcal{M}_a\right]^{2-(\alpha/a)} = \nu^{-\alpha/a} \mathcal{M}_a^{\frac{2a-\alpha}{a}}. \quad (2.23)$$

E:rhoMrep

3 Existence and uniqueness of the solution

The goal for this section is to prove Theorem 1.5. We first introduce some notation that will be used throughout the remainder of this paper. We start by defining $L_n(\vec{y})$ to be the Laplace transform of $\widetilde{f}_n(\vec{y}, 0, \cdot)$ evaluated at 1:

$$\begin{aligned} L_n(\vec{y}) &= n! \int_0^\infty e^{-t} \widetilde{f}_n(\vec{y}; 0, t) dt \\ &= \sum_{\sigma \in \Sigma_n} \int_0^\infty e^{-t} \int_{[0,t]^n} \prod_{k=1}^n G(s_k - s_{k-1}, y_{\sigma(k)} - y_{\sigma(k-1)}) d\vec{s} dt \end{aligned}$$

with the convention that. $s_0 = y_{\sigma(0)} = 0$. Next we define,

$$\begin{aligned} H_n(t, \vec{x}) &= n! \int_{(\mathbb{R}^d)^n} \prod_{k=1}^n K(x_k - y_k) \widetilde{f}_n(\vec{y}; 0, t) d\vec{y} \\ &= \sum_{\sigma \in \Sigma_n} \int_{[0,t]^n} \int_{(\mathbb{R}^d)^n} \prod_{k=1}^n K(x_k - y_k) \prod_{k=1}^n G(s_k - s_{k-1}, y_{\sigma(k)} - y_{\sigma(k-1)}) d\vec{y} d\vec{s} \end{aligned}$$

where in the last line we used relation (2.6) for expressing $\widetilde{f}_n(\cdot; 0, t)$. Next, similarly to L_n , we define $L(x)$ and can calculate $\mathcal{FL}(\xi)$ to be

$$L(x) = \int_0^\infty e^{-t} G(t, x) dt \quad \text{and} \quad \mathcal{FL}(\xi) = \int_0^\infty e^{-t} \mathcal{FG}(t, \cdot)(\xi) dt = \frac{1}{1 + \frac{\nu}{2} |\xi|^a} \quad (3.1)$$

E:LandFL

and through a change of variables, one sees that the Laplace transform of convolution in the time variable become a product:

$$L_n(\vec{y}) = \sum_{\sigma \in \Sigma_n} L(y_{\sigma(1)})L(y_{\sigma(2)} - y_{\sigma(1)}) \cdots L(y_{\sigma(n)} - y_{\sigma(n-1)}) \quad (3.2)$$

and using this and (3.1), we can calculate $\mathcal{F}L_n(\vec{\xi})$:

$$\mathcal{F}L_n(\vec{\xi}) = \sum_{\sigma \in \Sigma_n} \prod_{k=1}^n \frac{1}{1 + \frac{\nu}{2} \left| \sum_{j=k}^n \xi_j \right|^a}. \quad (3.3) \quad \boxed{\text{E:FLfn}}$$

L:Hneq1

Lemma 3.1. *Under the assumptions of Theorem 1.5, we have that*

$$\int_{(\mathbb{R}^d)^n} \left[\int_0^\infty e^{-t} H_n(t, \vec{x}) dt \right]^2 d\vec{x} = \int_{(\mathbb{R}^d)^n} |\mathcal{F}L_n(\vec{\xi})|^2 \mu(d\vec{\xi}). \quad (3.4)$$

Proof. Notice that

$$\begin{aligned} & \int_0^\infty e^{-t} H_n(t, \vec{x}) dt \\ &= \sum_{\sigma \in \Sigma_n} \int_0^\infty e^{-t} \int_{[0,t]^n} \int_{(\mathbb{R}^d)^n} \prod_{k=1}^n K(x_k - y_k) \prod_{k=1}^n G(s_k - s_{k-1}, y_{\sigma(k)} - y_{\sigma(k-1)}) d\vec{y} d\vec{s} dt \\ &= \int_{\mathbb{R}^{dn}} \prod_{k=1}^n K(x_k - y_k) L(\vec{y}) d\vec{y}. \end{aligned}$$

Therefore,

$$\begin{aligned} \int_{\mathbb{R}^{dn}} \left[\int_0^\infty e^{-t} H_n(t, \vec{x}) dt \right]^2 d\vec{x} &= \int_{\mathbb{R}^{dn}} \left[\int_{\mathbb{R}^{dn}} \prod_{k=1}^n K(x_k - y_k) L(\vec{y}) d\vec{y} \right]^2 d\vec{x} \\ &= \int_{\mathbb{R}^{dn}} |\mathcal{F}L_n(\vec{\xi})|^2 \prod_{k=1}^n \mu(d\xi_k) \end{aligned}$$

where the last line follows from Plancherel's theorem and the fact that $K * K = \gamma$. **consider adding the plancherel details** $\square$

L:inteq1

Lemma 3.2. *Under the assumptions of Theorem 1.5, we have that*

$$\left\| \widetilde{f}_n(\cdot, 0; t) \right\|_{\mathcal{H}^{\otimes n}}^2 = \frac{1}{(n!)^2} \int_{(\mathbb{R}^d)^n} H_n^2(t, \vec{x}) d\vec{x}. \quad (3.5)$$

Proof. Writing $\gamma(y_k - y'_k) = \int_{\mathbb{R}^d} K(y_k - x_k) K(y'_k - x_k) dx_k$, we see that

$$\begin{aligned} \left\| \widetilde{f}_n(\cdot, 0; t) \right\|_{\mathcal{H}^{\otimes n}}^2 &= \int_{(\mathbb{R}^d)^n} \int_{(\mathbb{R}^d)^n} \prod_{k=1}^n \gamma(y_k - y'_k) \widetilde{f}_n(\vec{y}, 0; t) \widetilde{f}_n(\vec{y}', 0; t) d\vec{y} d\vec{y}' \\ &= \int_{(\mathbb{R}^d)^n} \left[\int_{(\mathbb{R}^d)^n} \prod_{k=1}^n K(y_k - x_k) \widetilde{f}_n(\vec{y}, 0; t) d\vec{y} \right]^2 d(\vec{x}) \\ &= \int_{(\mathbb{R}^d)^n} \left[\frac{1}{n!} H_n(t, \vec{x}) \right]^2 d\vec{x}. \end{aligned}$$

maybe add the placherel theorem steps in the above equalities. $\square$

If f is a non-decreasing function on $[0, \infty)$, then the following reverse Cauchy-Schwarz inequality holds, which plays a key role in our developments.

L:intineq1

Lemma 3.3. *If $f : [0, \infty) \rightarrow [0, \infty)$ is a non-decreasing function, then*

$$2 \int_0^\infty e^{-2t} f^2(t) dt \leq \left(\int_0^\infty e^{-t} f(t) dt \right)^2. \quad (3.6)$$

Proof. Let τ and τ' be two independent random variables with exponential distribution of mean one. Then $\tau \wedge \tau'$ has an exponential distribution with density $g(t) = 2e^{-2t}$ and

$$\left(\int_0^\infty e^{-t} f(t) dt \right)^2 = \mathbb{E}[f(\tau)]\mathbb{E}[f(\tau')] = \mathbb{E}[f(\tau)f(\tau')] = 2 \int_0^\infty e^{-2t} f^2(t) dt.$$

□

Denote

$$T_n = T_n(\nu, a) := \int_{(\mathbb{R}^d)^n} \left[\sum_{\sigma \in \Sigma_n} \prod_{k=1}^n \frac{1}{1 + \frac{\nu}{2} \left| \sum_{j=k}^n \xi_{\sigma(j)} \right|^a} \right]^2 \mu(d\vec{\xi}). \quad (3.7)$$

E:Tn

By applying the substitution $\xi'_k = \left(\frac{\nu}{2}\right)^{1/a} \xi_k$ and the scaling property for φ , we may rewrite T_n as

$$T_n = \left(\frac{\nu}{2}\right)^{-n\alpha/a} T_n(2, a) \quad (3.8)$$

muassumption

Lemma 3.4. *Under the assumption of Theorem (1.8), we have that*

$$C_\mu = C_\mu(\nu, a) := \int_{(\mathbb{R}^d)} \left(\frac{1}{1 + \frac{\nu}{2} |\xi|^a} \right)^2 \mu(d\xi) < \infty \quad \text{and} \quad T_n \leq (n!)^2 (C_\mu)^2 \quad (3.9)$$

E:muassumpt

Proof. As in the proof of Lemma 3.6 [1], we apply a spherical substitution and get that

$$\int_{(\mathbb{R}^d)} \left(\frac{1}{1 + |\xi|^a} \right)^2 \mu(d\xi) = \left(\int_{S^{d-1}} \varphi(\gamma) d\sigma(\gamma) \right) \left(\int_0^\infty \frac{\rho^{\alpha-1}}{(1 + \rho^a)^2} d\rho \right).$$

The first integral is shown to be finite in [1]. The second integral is finite if

$$2a - \alpha + 1 > 1 \quad (3.10)$$

E:alpha2

which is already satisfied under the assumptions of Theorem 1.5. Hence the first relation in (3.9) is proven.

The second relation is proven the same was as in Lemma 3.6 [1]. Here is the proof.

$$\begin{aligned} T_n &\leq n! \int_{(\mathbb{R}^d)^n} \sum_{\sigma \in \Sigma_n} \left[\prod_{k=1}^n \frac{1}{1 + \frac{\nu}{2} \left| \sum_{j=k}^n \xi_{\sigma(j)} \right|^a} \right]^2 \mu(d\xi_1) \cdots \mu(d\xi_n) \\ &= (n!)^2 \int_{(\mathbb{R}^d)^n} \left[\prod_{k=1}^n \frac{1}{1 + \frac{\nu}{2} \left| \sum_{j=k}^n \xi_j \right|^a} \right]^2 \mu(d\xi_1) \cdots \mu(d\xi_n) \\ &\leq (n!)^2 \left[\sup_{\eta \in \mathbb{R}^d} \int_{(\mathbb{R}^d)^n} \prod_{k=1}^n \left(\frac{1}{1 + \frac{\nu}{2} |\xi + \eta|^a} \right)^2 \mu(d\xi) \right]^n \end{aligned}$$

$$\begin{aligned}
&= (n!)^2 \left[\int_{(\mathbb{R}^d)^n} \prod_{k=1}^n \left(\frac{1}{1 + \frac{\nu}{2} |\xi|^a} \right)^2 \mu(d\xi) \right]^n \\
&= (n!)^2 (C_\mu)^n
\end{aligned} \tag{3.11}$$

E:lem62step

where the last line follows by Lemma 4.1 of Balan and Song [2] or by Lemma 3.1 in [7]. $\square$

The following lemma shows that $\widetilde{f}_n(\cdot, 0; t) \in \mathcal{H}^{\otimes n}$ for any $t > 0$.

Lemma 3.5. *Under the assumptions of Theorem 1.5 and for any $t > 0$.*

$$\int_0^\infty e^{-t} \left\| \widetilde{f}_n(\cdot, 0; t) \right\|_{\mathcal{H}^{\otimes n}}^n dt \leq \frac{2^{n[2(b+r) - \frac{b\alpha}{a}]} T_n}{(n!)^2} \leq 2^{n[2(b+r) - \frac{b\alpha}{a}]} (C'_\mu)^n. \tag{3.12}$$

E:LNf9ineq

Proof. We first derive the following scaling property for $\mathcal{F}\widetilde{f}_n$ where $\widetilde{f}_n$ is the kernel of the n^{th} Wiener integral in the chaos expansion of the solution (2.2):

$$\mathcal{F}\widetilde{f}_n(\cdot, 0, ct)(\xi_1, \dots, \xi_n) = c^{n(b+r)} \mathcal{F}\widetilde{f}_n(\cdot, 0, t)(c^{b/a}\xi_1, \dots, c^{b/a}\xi_n). \tag{3.13}$$

E:Ffnscale

Let $c > 0$. Notice that

$$\mathcal{F}\widetilde{f}_n(\cdot, 0, ct)(\xi_1, \dots, \xi_n) = \frac{1}{n!} \sum_{\sigma \in \Sigma_n} \mathcal{F}f_n(\cdot, 0, ct)(\xi_{\sigma(1)}, \dots, \xi_{\sigma(n)}).$$

So it suffices to just look at the $\mathcal{F}f(\cdot, 0, ct)(\xi_{\sigma(1)}, \dots, \xi_{\sigma(n)})$ terms individually.

$$\begin{aligned}
\mathcal{F}f_n(\cdot, 0, ct)(\xi_1, \dots, \xi_n) &= \int_{[0, ct]_{<}^n} \prod_{k=1}^n \overline{\mathcal{F}G(t_{k+1} - t_k, \cdot)} \left(\sum_{j=1}^k \xi_j \right) d\vec{t} \\
&= \int_{[0, t]_{<}^n} \prod_{k=1}^n \overline{\mathcal{F}G(c(t_{k+1} - t_k), \cdot)} \left(\sum_{j=1}^k \xi_j \right) c^n d\vec{t}
\end{aligned}$$

where we applied the substitution $t'_i = t_i/c$ by applying the scaling property of $\mathcal{F}G$ we see that

$$\begin{aligned}
\int_{[0, t]_{<}^n} \prod_{k=1}^n \overline{\mathcal{F}G(c(t_{k+1} - t_k), \cdot)} \left(\sum_{j=1}^k \xi_j \right) c^n d\vec{t} &= \int_{[0, t]_{<}^n} \prod_{k=1}^n \overline{c^{b+r-1} \mathcal{F}G(t_{k+1} - t_k, \cdot)} \left(c^{b/a} \sum_{j=1}^k \xi_j \right) c^n d\vec{t} \\
&= c^{n(b+r)} \mathcal{F}f_n(\cdot, 0, t)(c^{b/a}\xi_1, \dots, c^{b/a}\xi_n).
\end{aligned}$$

Now by using this scaling property we see that

$$\begin{aligned}
\int_0^\infty e^{-t} \left\| \widetilde{f}_n(\cdot, 0, t) \right\|_{\mathcal{H}^{\otimes n}}^2 dt &= \int_0^\infty e^{-t} \int_{\mathbb{R}^{nd}} \left| \mathcal{F}\widetilde{f}_n(\cdot, 0; t)(\xi_1, \dots, \xi_n) \right|^2 \mu(d\vec{\xi}) dt \\
&= \int_0^\infty e^{-2t} \int_{\mathbb{R}^{nd}} \left| \mathcal{F}\widetilde{f}_n(\cdot, 0; 2t)(\xi_1, \dots, \xi_n) \right|^2 \mu(d\vec{\xi}) 2dt \\
&= 2^{2n(b+r)} \int_0^\infty 2e^{-2t} \int_{\mathbb{R}^{nd}} \left| \mathcal{F}\widetilde{f}_n(\cdot, 0, t)(2^{b/a}\xi_1, \dots, 2^{b/a}\xi_n) \right|^2 \mu(d\vec{\xi}) dt \\
&= 2^{2n(b+r)} \int_0^\infty 2e^{-2t} \int_{\mathbb{R}^{nd}} \left| \mathcal{F}\widetilde{f}_n(\cdot, 0; t)(\xi_1, \dots, \xi_n) \right|^2 2^{-\frac{nb\alpha}{a}} 2^{\frac{nb(d-\alpha)}{a}} \mu(d\vec{\xi}) dt \\
&= 2^{n(2(b+r) - b\alpha/a)} \int_0^\infty 2e^{-2t} \int_{\mathbb{R}^{nd}} \left| \mathcal{F}\widetilde{f}_n(\cdot, 0; t)(\xi_1, \dots, \xi_n) \right|^2 \mu(d\vec{\xi}) dt
\end{aligned}$$

$$= \frac{2^{n(2(b+r)-b\alpha/a)}}{(n!)^2} \int_0^\infty 2e^{-2t} \int_{\mathbb{R}^{nd}} H_n(t, \vec{x})^2 d\vec{x} dt$$

and now we apply Lemmas 3.2 and 3.3 as well as (3.3) to get

$$\begin{aligned} \frac{2^{n(2(b+r)-b\alpha/a)}}{(n!)^2} \int_0^\infty 2e^{-2t} \int_{\mathbb{R}^{nd}} H_n(t, \vec{x})^2 d\vec{x} dt &\leq \frac{2^{n(2(b+r)-b\alpha/a)}}{(n!)^2} \int_{\mathbb{R}^{nd}} \left(\int_0^\infty e^{-t} H_n(t, \vec{x}) dt \right)^2 d\vec{x} \\ &= \frac{2^{n(2(b+r)-b\alpha/a)}}{(n!)^2} \int_{\mathbb{R}^{nd}} |\mathcal{F}L_n(\vec{\xi})|^2 d\vec{\xi} \\ &= \frac{2^{n(2(b+r)-b\alpha/a)}}{(n!)^2} T_n, \end{aligned}$$

which proves the first inequality in (3.12). An application of Lemma 3.4 gives the second inequality in (3.12). $\square$

Proof of Theorem 1.5. First recall (3.13), which says

$$\mathcal{F}\widetilde{f}_n(\cdot, 0, ct)(\xi_1, \dots, \xi_n) = c^{n(b+r)} \mathcal{F}\widetilde{f}_n(\cdot, 0, t)(c^{b/a}\xi_1, \dots, c^{b/a}\xi_n).$$

In order to prove the theorem we need to show that $\sum_{n \geq 0} \theta^n n! \left\| \widetilde{f}_n(\cdot, 0, t) \right\|_{\mathcal{H}^{\otimes n}}^2 < \infty$. From the scaling property (2.9) we see that

$$\begin{aligned} \int_0^\infty e^{-t} \left\| \widetilde{f}_n(\cdot, 0, t) \right\|_{\mathcal{H}^{\otimes n}}^2 dt &= \left\| \widetilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2 \int_0^\infty e^{-t} t^{[2(b+r)-\frac{b\alpha}{a}]n} dt \\ &= \left\| \widetilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2 \Gamma \left(\left[2(b+r) - \frac{b\alpha}{a} \right] n + 1 \right) \end{aligned}$$

as long as we have

$$2(b+r) - \frac{b\alpha}{a} > 0. \quad (3.14) \quad \boxed{\text{E:alpha3}}$$

By using Lemma 3.5 we can deduce that

$$\begin{aligned} \left\| \widetilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2 &= \frac{\int_0^\infty e^{-t} \left\| \widetilde{f}_n(\cdot, 0, t) \right\|_{\mathcal{H}^{\otimes n}}^2 dt}{\Gamma \left(\left[2(b+r) - \frac{b\alpha}{a} \right] n + 1 \right)} \\ &\leq \frac{2^{n[2(b+r)-\frac{b\alpha}{a}]} T_n}{\Gamma \left(\left[2(b+r) - \frac{b\alpha}{a} \right] n + 1 \right)} \\ &= \frac{\left(2^{[2(b+r)-\frac{b\alpha}{a}]} C'_\mu \right)^n}{\Gamma \left(\left[2(b+r) - \frac{b\alpha}{a} \right] n + 1 \right)}. \end{aligned}$$

Hence, there exists a constant k_α depending on α such that

$$\left\| \widetilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2 \leq k_\alpha \frac{\left(2^{[2(b+r)-\frac{b\alpha}{a}]} C'_\mu \right)^n}{(n!)^{2(b+r)-b\alpha/a}}.$$

Therefore,

$$\sum_{n \geq 0} \theta^n n! \left\| \widetilde{f}_n(\cdot, 0, t) \right\|_{\mathcal{H}^{\otimes n}}^2 \leq \sum_{n \geq 0} \theta^n k_\alpha \frac{\left(2^{[2(b+r)-\frac{b\alpha}{a}]} C'_\mu \right)^n}{(n!)^{2(b+r)-1-b\alpha/a}},$$

where the sum is finite provided that

$$2(b+r) - 1 - b\alpha/a > 0. \quad (3.15) \quad \text{E:alpha4}$$

This completes the proof of Theorem 1.5. $\square$

4 Upper bound of the asymptotics

In this section we use the result obtained in Theorem 4.3 and the hypercontractivity property given by Theorem B.1 in [1] to prove the following theorem.

Theorem 4.1. *Under the assumptions of Theorem 1.5*

$$\limsup_{n \rightarrow \infty} t_p^{-\beta} \log \|u(t, x)\|_p \leq \left(\frac{1}{2}\right) \left(\frac{2a}{2a(b+r) - b\alpha}\right)^\beta (\theta\rho)^{\frac{a}{2a(b+r) - b\alpha - a}} \left(2(b+r) - \frac{b\alpha}{a} - 1\right).$$

In order to prove this theorem, we need to first prove the case when $p = 2$ (Lemma 4.3 below) and then use the hypercontractivity to establish the general case.

Lemma 4.2. *Under the assumptions of Theorem 1.5, we have that*

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \left(\left\| \widetilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2 \Gamma \left(\left[2(b+r) - \frac{b\alpha}{a} \right] n + 1 \right) \right) = \log \left(2^{2(b+r) - \frac{b\alpha}{a}} \rho \right) \quad (4.1) \quad \text{E:2ineq-a}$$

and

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \left((n!)^{2(b+r) - \frac{b\alpha}{a}} \left\| \widetilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2 \right) = \log \left(\left(\frac{2}{2(b+r) - \frac{b\alpha}{a}} \right)^{2(b+r) - \frac{b\alpha}{a}} \right) + \log \rho. \quad (4.2) \quad \text{E:2ineq-b}$$

Proof. Recall the definition of T_n from above as well as (2.16) which says that

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \left[\frac{1}{(n!)^2} \int_{(\mathbb{R}^d)^n} \left(\sum_{\sigma \in \Sigma_n} \prod_{k=1}^n \frac{1}{1 + \frac{\nu}{2} |\sum_{j=k}^n \xi_{\sigma(j)}|^a} \right)^2 \mu(d\vec{\xi}) \right] = \lim_{n \rightarrow \infty} \frac{1}{n} \log \left[\frac{T_n}{(n!)^2} \right] = \log \rho.$$

We already showed in the proof of Theorem 1.5 above that

$$\int_0^\infty e^{-t} \left\| \widetilde{f}_n(\cdot, 0, t) \right\|_{\mathcal{H}^{\otimes n}}^2 dt = \left\| \widetilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2 \Gamma \left(\left[2(b+r) - \frac{b\alpha}{a} \right] n + 1 \right)$$

and then Lemma 3.5 above shows that

$$\left\| \widetilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2 \Gamma \left(\left[2(b+r) - \frac{b\alpha}{a} \right] n + 1 \right) \leq \frac{2^{n \left[2(b+r) - \frac{b\alpha}{a} \right]}}{(n!)^2} T_n.$$

Combining this and (2.16) we see that

$$\begin{aligned} \limsup_{n \rightarrow \infty} \frac{1}{n} \log & \left[\left\| \widetilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2 \Gamma \left(\left[2(b+r) - \frac{b\alpha}{a} \right] n + 1 \right) \right] \\ & \leq \log \left(2^{\left[2(b+r) - \frac{b\alpha}{a} \right]} \right) + \lim_{n \rightarrow \infty} \frac{1}{n} \log \left(\frac{T_n}{(n!)^2} \right) \\ & = \log \left(2^{\left[2(b+r) - \frac{b\alpha}{a} \right]} \right) + \log(\rho) \end{aligned}$$

which proves the upper bound for (4.1).

Now we prove the lower bound. Define for $t > 0$ and $t' > 0$

$$J_n(t, t') = \int_{\mathbb{R}^{nd}} H_n(t, x) H_n(t', x) dx$$

and recall Lemma 3.2 which says

$$\left\| \widetilde{f}_n(\cdot, 0; t) \right\|_{\mathcal{H}^{\otimes n}}^2 = \frac{1}{(n!)^2} \int_{(\mathbb{R}^d)^n} H_n(t, \vec{x})^2 d\vec{x} = \frac{1}{(n!)^2} J_n(t, t).$$

Thus $J_n(t, t)$ satisfies the scaling property

$$J_n(t, t) = t^{[2(b+r)-b\alpha/a]n} J_n(1, 1).$$

Let τ and $\tilde{\tau}$ be independent exponential random variables with mean 1. By lemma 3.1,

$$\begin{aligned} \mathbb{E}[J_n(\tau, \tau')] &= \int_0^\infty \int_0^\infty e^{-t} e^{-\tilde{t}} J_n(t, \tilde{t}) dt d\tilde{t} = \int_{\mathbb{R}^{dn}} \left[\int_0^\infty e^{-t} H_n(t, x) dt \right]^2 dx \\ &= \int_{\mathbb{R}^{nd}} |\mathcal{F}L_n(\xi_1, \dots, \xi_n)|^2 \mu(d\xi_1) \cdots \mu(d\xi_n) =: T_n. \end{aligned}$$

In addition,

$$J_n(t, t') \leq J_n(t, t)^{1/2} J_n(t', t')^{1/2} = t^{[2(b+r)-b\alpha/a]n/2} t'^{[2(b+r)-b\alpha/a]n/2} J_n(1, 1).$$

Combining all of this gives us

$$\begin{aligned} T_n &= \mathbb{E}[J_n(\tau, \tilde{\tau})] \leq \mathbb{E} \left[\tau^{[2(b+r)-b\alpha/a]n/2} \right] \mathbb{E} \left[\tilde{\tau}^{[2(b+r)-b\alpha/a]n/2} \right] J_n(1, 1) \\ &= \Gamma \left(\frac{[2(b+r)-b\alpha/a]n}{2} \right)^2 J_n(1, 1) \\ &= \Gamma \left(\frac{[2(b+r)-b\alpha/a]n}{2} \right)^2 (n!)^2 \left\| \widetilde{f}_n(\cdot, 0; 1) \right\|_{\mathcal{H}^{\otimes n}}^2, \end{aligned}$$

where the last line follows from Lemma 3.2. Thus so far we have shown

$$\frac{T_n}{(n!)^2} = \frac{\left(\frac{\nu}{2}\right)^{-n\alpha/a} T'_n}{(n!)^2} \leq \Gamma \left(\frac{[2(b+r)-b\alpha/a]n}{2} \right)^2 \left\| \widetilde{f}_n(\cdot, 0; 1) \right\|_{\mathcal{H}^{\otimes n}}^2. \quad (4.3) \quad \boxed{\text{E:2lowbd}}$$

Now, another application of Stirling's formula shows that

$$\Gamma \left(\frac{[2(b+r)-b\alpha/a]n}{2} \right)^2 \sim \Gamma([2(b+r)-b\alpha/a]n+1) 2^{-[2(b+r)-b\alpha/a]n} C_n$$

where $C_n = 2^{-1}[2(b+r)-b\alpha/a]^{1/2}(2\pi n)^{1/2}$. We now infer from (4.3) that

$$c_\alpha \frac{2^{[2(b+r)-b\alpha/a]n}}{C_n} \frac{1}{(n!)^2} T_n \leq \Gamma([2(b+r)-b\alpha/a]n+1) \left\| \widetilde{f}_n(\cdot, 0; 1) \right\|_{\mathcal{H}^{\otimes n}}^2$$

where c_α is a constant depending on α . Since $\frac{1}{n} \log(C_n) \rightarrow 0$ as $n \rightarrow \infty$, we obtain the lower bound

$$\liminf_{n \rightarrow \infty} \frac{1}{n} \log \left(\Gamma([2(b+r)-b\alpha/a]n+1) \left\| \widetilde{f}_n(\cdot, 0; 1) \right\|_{\mathcal{H}^{\otimes n}}^2 \right)$$

$$\geq \log \left(2^{[2(b+r)-b\alpha/a]} \right) + \log(\rho)$$

Thus (4.1) is proven. Lastly, (4.2) follows from (4.1) and the fact that

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \left[\frac{\Gamma(an+1)}{(n!)^a} \right] = a \log(a).$$

This completes the proof of Lemma 4.2. □

L:p=2

Lemma 4.3. *Under the assumptions of Theorem 1.5, we have that*

$$\lim_{t \rightarrow \infty} \frac{1}{t^\beta} \log \mathbb{E}(|u(t, x)|^2) = \left(\frac{2a}{2a(b+r) - b\alpha} \right)^\beta (\theta \rho)^{\frac{a}{2a(b+r) - b\alpha - a}} \left(2(b+r) - \frac{b\alpha}{a} - 1 \right).$$

Proof. Assuming that there is an $L^2(\Omega)$ solution $u(t, x)$ and then applying the scaling property (2.9) gives us

$$\begin{aligned} \mathbb{E}(|u(t, x)|^2) &= \sum_{n \geq 0} \theta^n (n!) \left\| \tilde{f}_n(\cdot, 0, t) \right\|_{\mathcal{H}^{\otimes n}}^2 = \sum_{n \geq 0} \theta^n (n!) t^{(2(b+r) - b\alpha/a)n} \left\| \tilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2 \\ &= \sum_{n \geq 0} x_n R_n t^{(2(b+r) - b\alpha/a)n} \end{aligned}$$

where

$$R_n = (n!)^{2(b+r) - b\alpha/a} \left\| \tilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2 \quad \text{and} \quad x_n = \frac{\theta^n}{(n!)^{2(b+r) - (b\alpha/a) - 1}}.$$

Notice that (4.2) above says that

$$\frac{1}{n} \log(R_n) \rightarrow \log \left(\left(\frac{2}{2(b+r) - \frac{b\alpha}{a}} \right)^{2(b+r) - \frac{b\alpha}{a}} \rho \right) \quad \text{as } n \rightarrow \infty.$$

Now define R to be

$$R = \left(\frac{2}{2(b+r) - \frac{b\alpha}{a}} \right)^{2(b+r) - \frac{b\alpha}{a}} \rho.$$

We want to find a β and A so that

$$\lim_{t \rightarrow \infty} \frac{1}{t^\beta} \log \sum_{n \geq 0} x_n R^n \left(t^{(2(b+r) - b\alpha/a)} \right)^n = A$$

By Lemma A.3 of [1],

$$\lim_{t \rightarrow \infty} \left[\frac{1}{(\theta R) t^{2(b+r) - b\alpha/a}} \right]^{\frac{1}{2(b+r) - (b\alpha/a) - 1}} \log \sum_{n \geq 0} \frac{[(\theta R) t^{2(b+r) - b\alpha/a}]^n}{(n!)^{2(b+r) - (b\alpha/a) - 1}} = 2(b+r) - (b\alpha/a) - 1$$

and by an easy algebraic manipulation

$$\lim_{t \rightarrow \infty} \left[\frac{1}{t^{2(b+r) - b\alpha/a}} \right]^{\frac{1}{2(b+r) - (b\alpha/a) - 1}} \log \sum_{n \geq 0} \frac{[(\theta R) t^{2(b+r) - b\alpha/a}]^n}{(n!)^{2(b+r) - (b\alpha/a) - 1}}$$

$$= [2(b+r) - (b\alpha/a) - 1](\theta R)^{\frac{1}{2(b+r) - (b\alpha/a) - 1}}.$$

Hence,

$$\beta = \frac{2a(b+r) - b\alpha}{2a(b+r) - (b\alpha) - a} \quad \text{and} \quad A = [2(b+r) - (b\alpha/a) - 1](\theta R)^{\frac{a}{2a(b+r) - b\alpha - a}}.$$

Finally, an application of Lemma A.2 of [1] shows that

$$\begin{aligned} & \lim_{t \rightarrow \infty} \frac{1}{t^\beta} \log \mathbb{E}(|u(t, x)|^2) \\ &= \theta^{\frac{a}{2a(b+r) - b\alpha - a}} \left(\frac{2a}{2a(b+r) - b\alpha} \right)^\beta \rho^{\frac{a}{2a(b+r) - b\alpha - a}} \left(2(b+r) - \frac{b\alpha}{a} - 1 \right), \end{aligned}$$

which completes the proof of Theorem 4.3. $\square$

Now we are ready to prove Theorem 4.1.

Proof of Theorem 4.1. We denote u_λ for $\lambda > 0$ to be the solution of the fractional equation with θ replaced with λ and $u = u_\theta$. Using theorem B.1 of [1], we see that for all $p \geq 2$,

$$\|u(t, 0)\|_p \leq \|u_{(p-1)\theta}(t, 0)\|_2$$

and by the scaling property

$$\begin{aligned} \|u_{(p-1)\theta}(t, 0)\|_2^2 &= \sum_{n \geq 0} (n!)^n \theta^n (p-1)^n \left\| \tilde{f}_n(\cdot, 0, t) \right\|_{\mathcal{H}^{\otimes n}}^2 \\ &= \sum_{n \geq 0} (n!)^n \theta^n \left\| \tilde{f}_n\left(\cdot, 0, t(p-1)^{\frac{1}{2(b+r) - \frac{1}{b\alpha/a}}}\right) \right\|_{\mathcal{H}^{\otimes n}}^2 \\ &= \left\| u\left(t(p-1)^{\frac{1}{2(b+r) - \frac{1}{b\alpha/a}}}, 0\right) \right\|_2^2. \end{aligned}$$

Hence,

$$\|u(t, 0)\|_p \leq \left\| u\left(t(p-1)^{\frac{1}{2(b+r) - \frac{1}{b\alpha/a}}}, 0\right) \right\|_2 =: \|u(t_p, 0)\|_2$$

and an application of Lemma 4.3 proves Theorem 4.1. $\square$

5 Lower bound of the asymptotics

The goal of this section is to prove the following theorem.

Theorem 5.1. *Under the assumptions of Theorem 1.5*

$$\liminf_{n \rightarrow \infty} t_p^{-\beta} \log \|u(t, x)\|_p \geq \left(\frac{1}{2}\right) \left(\frac{2a}{2a(b+r) - b\alpha}\right)^\beta (\theta \rho)^{\frac{a}{2a(b+r) - b\alpha - a}} \left(2(b+r) - \frac{b\alpha}{a} - 1\right).$$

We will need to build up some theory before being able to prove the above theorem. We start by defining the function $W_n(t, \phi)$ on $(0, \infty) \times L_{\mathbb{C}}^2(\mu)$ by

$$W_n(t, \phi) := \int_{[0, t]^n} \int_{(\mathbb{R}^d)^n} \prod_{k=1}^n \phi(\xi_k) \prod_{k=1}^n \mathcal{F}G(s_k - s_{k-1}, \cdot)(\xi_k + \dots + \xi_n) \mu(d\xi_1) \dots \mu(d\xi_n) ds \quad (5.1)$$

with $s_0 = 0$. We now give conditions for when W_n is well defined.

L:LWn

Lemma 5.2. *If the measure μ satisfies the relation in (3.9), then $W_n(t, \phi)$ is well defined and for any $d \geq 1$, $t > 0$ and $\phi \in L_{\mathbb{C}}^2(\mu)$. Moreover,*

$$\int_0^\infty e^{-t} W_n(t, \phi) dt = \int_{(\mathbb{R}^d)^n} \prod_{k=1}^n \phi(\xi_k) \prod_{k=1}^n \frac{1}{1 + \frac{\nu}{2} |\xi_k + \dots + \xi_n|^a} \mu(d\xi) \cdots \mu(d\xi_n). \quad (5.2)$$

E:LWn

Proof. The proof follows the exact arguments as those in the proof of Lemma 6.2 of [1] except that one needs to use the following Laplace transform:

$$\int_0^\infty e^{-t} \mathcal{F}G(t, \cdot)(\xi) dt = \frac{1}{1 + \frac{\nu}{2} |\xi|^a},$$

which is due to ... □

The next Proposition is a restatement of Proposition 6.3 of [1]. The proof follows the same proof as that of Proposition 6.3 of [1]. We will not repeat here.

P:Wnprop1

Proposition 5.3. *For $f \in \mathcal{H}$, $t > 0$, and $p, q > 1$ with $p^{-1} + q^{-1} = 1$, it holds that*

$$\|u(t, 0)\|_p \geq \exp \left\{ -\frac{1}{2} (q-1) \|f\|_{\mathcal{H}}^2 \right\} \left| \sum_{n \geq 0} \theta^{n/2} W_n(t, \mathcal{F}f) \right| \quad (5.3)$$

E:Wnprop1

and as a consequence, the series $\left| \sum_{n \geq 0} \theta^{n/2} W_n(t, \mathcal{F}f) \right|$ converges provided that $\|u(t, 0)\|_p < \infty$, which is the case under Theorem 1.5.

Now we are going to apply a scaling argument to (5.3) in order to put t and p together, from which we can determine the constants t_p and β defined in (1.9).

P:Wnprop2

Proposition 5.4. *For $p, q > 1$ and $p^{-1} + q^{-1} = 1$ we have that*

$$\|u(t, 0)\|_p \geq \exp \left\{ -\frac{1}{2} t_p^\beta \|f\|_{\mathcal{H}}^2 \right\} \left| \sum_{n \geq 0} \theta^{n/2} W_n \left(t_p^\beta, \mathcal{F}f \right) \right|, \quad (5.4)$$

E:Wnprop2

where the constants β and t_p are defined in (1.9).

Proof. From Proposition 5.3, we see that for any $f \in \mathcal{H}$, the inequality (5.3) holds. For some constants $V, W > 0$, which will be determined in this proof, let $f_\tau(x) := \tau^V f(\tau^W x)$ be a scaled version of f . It is clear that $f_\tau \in \mathcal{H}$. By some elementary scaling arguments (see the proof of the Lemma 6.4 of [1]), one can show that

$$\|f_\tau\|_{\mathcal{H}}^2 = \tau^{2(V-dW)+W\alpha} \|f\|_{\mathcal{H}}^2 \quad \text{and} \quad (5.5)$$

$$W_n(t, \mathcal{F}f_\tau) = \tau^{n[V-W((d-\alpha)+\frac{a}{b}(b+r))]} W_n \left(t \tau^{\frac{a}{b}W}, \mathcal{F}f \right). \quad (5.6)$$

Hence, an application of Proposition 5.3 to f_τ shows that

$$\begin{aligned} \|u(t, 0)\|_p &\geq \exp \left\{ -\frac{1}{2} (q-1) \|f_\tau\|_{\mathcal{H}}^2 \right\} \left| \sum_{n \geq 0} \theta^{n/2} W_n(t, \mathcal{F}f_\tau) \right| \\ &= \exp \left\{ -\frac{1}{2} (q-1) \tau^{2(V-dW)+W\alpha} \|f\|_{\mathcal{H}}^2 \right\} \left| \sum_{n \geq 0} \theta^{n/2} \tau^{n[V-W((d-\alpha)+\frac{a}{b}(b+r))]} W_n \left(t \tau^{\frac{a}{b}W}, \mathcal{F}f \right) \right|. \end{aligned}$$

Comparing the above lower bound with that of (5.4), we obtain the following two relations with three unknowns W , V and τ :

$$V - W \left((d - \alpha) + \frac{a}{b}(b + r) \right) = 0, \quad (5.7) \quad \boxed{\text{E:VW-1}}$$

$$(p - 1)^{-1} \tau^{2(V - dW) + W\alpha} = t \tau^{\frac{a}{b}W}. \quad (5.8) \quad \boxed{\text{E:VW-2}}$$

Since (5.8) should hold for all $t > 0$ and $p \geq 2$, one can choose $\tau = (p - 1)t$ to reduce the relation (5.8) to

$$\tau^{2(V - dW) + W\alpha} = \tau^{\frac{a}{b}W + 1},$$

which then gives the following equation

$$2(V - dW) + W\alpha = 1 + \frac{a}{b}W. \quad (5.9) \quad \boxed{\text{E:VW-3}}$$

Now solve the linear equations (5.7) and (5.9) for W and V to see that

$$\begin{cases} W = \frac{b}{a}(\beta - 1), \\ V = \left(\frac{a}{b}(b + r) - \alpha + d \right) \frac{b}{a}(\beta - 1), \end{cases} \quad \text{with } \beta := \frac{2(b + r) - \frac{b\alpha}{a}}{2(b + r) - \frac{b\alpha}{a} - 1}.$$

Therefore, the scaling f_τ and t_p^β should be

$$f_\tau(x) = \tau^{\left(\frac{a}{b}(b+r)-\alpha+d\right)\frac{b}{a}(\beta-1)} f\left(\tau^{\frac{b}{a}(\beta-1)}x\right) \quad \text{and} \quad t_p^\beta := t \tau^{\frac{a}{b}W} = (p - 1)^{\beta-1} t^\beta,$$

respectively. This completes the proof of Proposition 5.4. $\square$

To remove the absolute value sign in Proposition 5.4, we want to identify all the $f \in \mathcal{H}$ for which $W_n(t, \mathcal{F}f)$ is nonnegative. In fact, if we consider the space

$$\mathcal{H}_+ = \{f \in \mathcal{H} : f \text{ is nonnegative and nonnegative definite}\}$$

then by Plancherel's theorem

$$W_n(t, \mathcal{F}f) = \int_{[0,t]^n} \int_{\mathbb{R}^{nd}} \prod_{k=1}^n (f * \gamma)(x_k) \prod_{k=1}^n G(s_k - s_{k-1}, x_k - x_{k-1}) d\vec{x} d\vec{s} =: U_n(t, f)$$

with $s_0 = 0$ and $x_0 = 0$. Under Assumption 1.4, the Green's function for the fractional equation is nonnegative and hence under these conditions we get that $W_n(t, \mathcal{F}(f)) \geq 0$ for all the functions $f \in \mathcal{H}_+$. Hence we have that

$$0 \leq W_n(t, \mathcal{F}f) = U_n(t, f) < \infty \quad \text{for all } f \in \mathcal{H}_+ \text{ and for any of the four options in Assumption 1.4.}$$

where the finiteness comes from Proposition 5.4.

Now we define

$$W_n(t) := \sup_{f \in \mathcal{H}, \|f\|_{\mathcal{H}}=1} W_n(t, \mathcal{F}(f)) \quad \text{and} \quad U_n(t) := \sup_{f \in \mathcal{H}_+, \|f\|_{\mathcal{H}}=1} U_n(t, f)$$

and with this we see that

$$W_n(t) \geq U_n(t) \geq 0$$

Proposition 5.5. *If τ is an exponential random variable with mean one, then*

$$\liminf_{n \rightarrow \infty} \frac{1}{n} \log \mathbb{E}(U_n(\tau)) \geq \log(\rho^{1/2}).$$

Proof. We start by letting τ be an exponential random variable with mean one. With this,

$$\mathbb{E}(U_n(\tau)) = \int_0^\infty e^{-t} U_n(t) dt \geq \sup_{f \in \mathcal{H}_+, \|f\|_{\mathcal{H}}=1} \int_0^\infty e^{-t} U_n(t, f) dt$$

For an $f \in \mathcal{H}_+$ and $\|f\|_{\mathcal{H}} = 1$ we have that $\mathcal{F}f$ is nonnegative and nonnegative definite which further implies that $\mathcal{F}f$ is even. In addition, Lemma 5.2 gives us that

$$\int_0^\infty e^{-t} U_n(t, f) dt = \int_0^\infty e^{-t} W_n(t, \mathcal{F}f) dt = \int_{\mathbb{R}^{dn}} \prod_{k=1}^n \mathcal{F}f(\xi_k) \prod_{k=1}^n \frac{1}{1 + \frac{\nu}{2} |\xi_k + \dots + \xi_n|^a} \mu(d\xi_1) \dots \mu(d\xi_n)$$

Modifying the ideas in the proof of Theorem 1.2 [3] gives us the following identity:

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \int_{\mathbb{R}^{dn}} \prod_{k=1}^n (\mathcal{F}f)(\xi_k) \prod_{k=1}^n \frac{1}{1 + \frac{\nu}{2} |\xi_k + \dots + \xi_n|^a} \mu(d\xi_1) \dots \mu(d\xi_n) = \log[\rho(\mathcal{F}f)]$$

where $\rho(\cdot)$ is defined on the space of nonnegative and nonnegative definite functions and is given by

$$\rho(g) := \sup_{\|h\|_{L^2}=1} \int_{\mathbb{R}^d} g(\xi) \left[\int_{\mathbb{R}^d} \frac{h(\xi + \eta) h(\eta)}{\sqrt{1 + \frac{\nu}{2} |\xi + \eta|^a} \sqrt{1 + \frac{\nu}{2} |\eta|^a}} d\eta \right] \mu(d\xi).$$

Note that since g is nonnegative and since μ has a nonnegative density φ then we may assume h is also nonnegative for the remainder of this proof. With this being said, we see that for any $f \in \mathcal{H}_+$ and $\|f\|_{\mathcal{H}} = 1$

$$\liminf_{n \rightarrow \infty} \frac{1}{n} \log \mathbb{E}(U_n(t)) \geq \log \rho(\mathcal{F}f)$$

We now need to calculate

$$\sup_{f \in \mathcal{H}_+, \|f\|_{\mathcal{H}}=1} \rho(\mathcal{F}f)$$

Consider a nonnegative function $h \in L^2(\mathbb{R}^d)$. The function $h(\cdot)/\sqrt{1 + \frac{\nu}{2} |\cdot|^a} \in L^2(\mathbb{R}^d)$ so that $g_h(x) := (2\pi)^{d/2} \mathcal{F}^{-1} \left(\frac{h(\cdot)}{\sqrt{1 + \frac{\nu}{2} |\cdot|^a}} \right) (x)$ is well defined. **I still need to verify the details that** under these conditions, $g \in W_{1,a}(\mathbb{R}^d)$ with $\|g\|_{W_{1,a}} = 1$. Here is a sketch of the details. **we no longer need this part since we do not need g to be continuous, only need to be nonnegative and nonnegative definite.**

$$\begin{aligned} \|g\|_{W_{1,a}}^2 &= \|g\|_{L^2}^2 + (2\pi)^{-d} \sum_{k=1}^d \int_{\mathbb{R}^d} \xi_k^a |\mathcal{F}g(\xi)|^2 d\xi \\ &= (2\pi)^{-d} \int_{\mathbb{R}^d} (1 + |\xi|^a) |\mathcal{F}g(\xi)|^2 d\xi \end{aligned}$$

$$= \int_{\mathbb{R}^d} h(\xi)^2 d\xi = 1$$

where above we used the fact that $\|g\|_{L^2}^2 = (2\pi)^{-d} \|\mathcal{F}g\|_{L^2}^2$.

We need from this point on.

Now, for notation let $\tilde{h}(x) = h(-x)$. Note that

$$\begin{aligned} \int_{\mathbb{R}^d} \frac{h(\xi + \eta)h(\eta)}{\sqrt{1 + \frac{\nu}{2}|\xi + \eta|^a} \sqrt{1 + \frac{\nu}{2}|\eta|^a}} d\eta &= (2\pi)^{-d} \int_{\mathbb{R}^d} \mathcal{F}g_h(\xi - \eta) \mathcal{F}g_h(-\eta) d\eta \\ &= (2\pi)^{-d} (\mathcal{F}g_h * \widetilde{\mathcal{F}g_h})(\xi) \end{aligned}$$

Since $h(\cdot)$ is real valued then $\widetilde{\mathcal{F}g_h} = \overline{\mathcal{F}g_h} = \mathcal{F}\bar{g}_h$. Using this and the fact that $\mathcal{F}(fg) = (2\pi)^{-d} \mathcal{F}(f) * \mathcal{F}(g)$, we see that $(2\pi)^{-d} (\mathcal{F}g_h * \widetilde{\mathcal{F}g_h})(\xi) = \mathcal{F}[|g_h|^2](\xi)$ and with this we see that

$$\begin{aligned} \rho(\mathcal{F}f) &= \sup_{\|h\|_{L^2}=1} (2\pi)^{-d} \int_{\mathbb{R}^d} \mathcal{F}f(\xi) (\mathcal{F}g_h * \widetilde{\mathcal{F}g_h})(\xi) \mu(d\xi) \\ &= \sup_{\|h\|_{L^2}=1} \int_{\mathbb{R}^d} \mathcal{F}f(\xi) \mathcal{F}[|g_h|^2](\xi) \mu(d\xi) \\ &= \sup_{\|h\|_{L^2}=1} \langle f, |g_h|^2 \rangle_{\mathcal{H}}. \end{aligned}$$

With this we see that

$$\sup_{f \in \mathcal{H}, \|f\|_{\mathcal{H}}=1} \rho(\mathcal{F}f) = \sup_{f \in \mathcal{H}, \|f\|_{\mathcal{H}}=1} \sup_{\|h\|_{L^2}=1} \langle f, |g_h|^2 \rangle_{\mathcal{H}} = \sup_{\|h\|_{L^2}=1} \langle |g_h|^2, |g_h|^2 \rangle_{\mathcal{H}}^{1/2}$$

where the optimal f is chosen to be $|g_h|^2 / \| |g_h|^2 \|_{\mathcal{H}}$. Now we want to show that

$$|g_h|^2 \in \mathcal{H}_+ \tag{5.10} \quad \boxed{\text{E:GH+}}$$

which would imply that the supremum can be restricted to $\mathcal{H}_+$. If this were the case then we would have that

$$\sup_{f \in \mathcal{H}_+, \|f\|_{\mathcal{H}}=1} \rho(\mathcal{F}f) = \sup_{\|h\|_{L^2}=1} \langle |g_h|^2, |g_h|^2 \rangle_{\mathcal{H}}^{1/2}$$

and notice that

$$\begin{aligned} \langle |g_h|^2, |g_h|^2 \rangle_{\mathcal{H}}^{1/2} &= \left(\int_{\mathbb{R}^d} |\mathcal{F}[|g_h|^2](\xi)|^2 \mu(d\xi) \right)^{1/2} \\ &= (2\pi)^{-d} \left(\int_{\mathbb{R}^d} |(\mathcal{F}g_h * \widetilde{\mathcal{F}g_h})(\xi)|^2 \mu(d\xi) \right)^{1/2} \\ &= \left(\int_{\mathbb{R}^d} \left[\int_{\mathbb{R}^d} \frac{h(\xi + \eta)h(\eta)}{\sqrt{1 + \frac{\nu}{2}|\xi + \eta|^a} \sqrt{1 + \frac{\nu}{2}|\eta|^a}} d\eta \right]^2 \mu(d\xi) \right)^{1/2} \\ &= \rho^{1/2} \end{aligned}$$

which finally leads us to the equality

$$\liminf_{n \rightarrow \infty} \frac{1}{n} \log \mathbb{E}(U_n(\tau)) \geq \log(\rho^{1/2}).$$

Hence, we finish the proof by proving (5.10). First note that from the above we see that

$$\int_{\mathbb{R}^d} |\mathcal{F}|g_h|^2(\xi)|^2 \mu(d\xi) \leq \rho < \infty$$

and so $|g_h|^2 \in \mathcal{H}$. Lastly, since $\mathcal{F}|g_h|^2(\xi)$ is nonnegative for $\xi \in \mathbb{R}^d$ and for nonnegative $h \in L^2(\mathbb{R}^d)$ then $\nu(d\xi) = \mathcal{F}|g_h|^2(\xi)d\xi$ is a nonnegative tempered measure. The Bochner-Schwarz theorem then implies that $|g_h|^2$ is nonnegative and nonnegative definite. This shows that $|g_h|^2 \in \mathcal{H}_+$. This completes the proof of Proposition 5.5. $\square$

L:LogU

Lemma 5.6. *It holds that*

$$\liminf_{n \rightarrow \infty} \frac{1}{n} \log \left[(n!)^{\frac{b}{a}(a - \frac{\alpha}{2} + \frac{a}{b}r)} U_n(1) \right] \geq \log \left(\frac{\rho^{1/2}}{\left(\frac{b}{a}(a - \frac{\alpha}{2} + \frac{a}{b}r) \right)^{\frac{b}{a}(a - \frac{\alpha}{2} + \frac{a}{b}r)}} \right). \quad (5.11) \quad \text{E:LogU}$$

Proof. Notice that by some elementary scaling arguments, we have that for all $t > 0$,

$$W_n(t) = t^{n \frac{b}{a}(a - \frac{\alpha}{2} + \frac{a}{b}r)} W_n(1) \quad \text{and} \quad U_n(t) = t^{n \frac{b}{a}(a - \frac{\alpha}{2} + \frac{a}{b}r)} U_n(1). \quad (5.12) \quad \text{E:WU}$$

Let τ be an exponential random variable with mean one. By (5.12),

$$\mathbb{E}[U_n(\tau)] = \mathbb{E} \left(\tau^{n \frac{b}{a}(a - \frac{\alpha}{2} + \frac{a}{b}r)} U_n(1) \right) = \Gamma \left(n \frac{b}{a} \left(a - \frac{\alpha}{2} + \frac{a}{b}r \right) + 1 \right) U_n(1).$$

Then an application of Proposition 5.5 shows that

$$\liminf_{n \rightarrow \infty} \frac{1}{n} \log \left[\Gamma \left(n \frac{b}{a} \left(a - \frac{\alpha}{2} + \frac{a}{b}r \right) + 1 \right) U_n(1) \right] \geq \log(\rho^{1/2}). \quad (5.13) \quad \text{E:1b1}$$

As a direct consequence of Stirling's formula, we have that

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \left(\frac{\Gamma(an + 1)}{(n!)^a} \right) = a \log(a), \quad \text{for all } a > 0,$$

with which one immediately sees that

$$\liminf_{n \rightarrow \infty} \frac{1}{n} \log \left(\frac{\Gamma \left(n \frac{b}{a} \left(a - \frac{\alpha}{2} + \frac{a}{b}r \right) + 1 \right)}{(n!)^{\frac{b}{a}(a - \frac{\alpha}{2} + \frac{a}{b}r)}} \right) = \frac{b}{a} \left(a - \frac{\alpha}{2} + \frac{a}{b}r \right) \log \left(\frac{b}{a} \left(a - \frac{\alpha}{2} + \frac{a}{b}r \right) \right),$$

where the condition

$$\frac{b}{a} \left(a - \frac{\alpha}{2} + \frac{a}{b}r \right) > 0$$

is guaranteed by (3.15) (or (1.7)). Finally, (5.11) is proven after multiplying and dividing $(n!)^{\frac{b}{a}(a - \frac{\alpha}{2} + \frac{a}{b}r)}$ on the inside of the log in (5.13). $\square$

L:logUnlb

Lemma 5.7. *For any $k, \theta > 0$, there exists a constant $c_1 = c_1(\alpha, \mathcal{M}, k, \theta) > 0$ such that, by setting $n_t = \lfloor c_1 t \rfloor$, it holds that*

$$\liminf_{t \rightarrow \infty} \frac{1}{t} \log(k^{n_t} \theta^{n_t/2} U_{n_t}(t)) \geq \left(k\sqrt{\theta} \right)^{\frac{2a}{2ab - \alpha b + 2ar}} \left(\rho^{1/2} \right)^{\frac{2a}{2ab - \alpha b + 2ar}}. \quad (5.14) \quad \text{E:69ineq}$$

Proof. Let $\epsilon > 0$. Lemma 5.6 guarantees the existence of an $N_\epsilon > 0$ so for all $n > N_\epsilon$

$$(n!)^{\frac{b}{a}(a-\frac{\alpha}{2}+\frac{a}{b}r)} U_n(1) \geq \exp(n(\log(R) - \epsilon)) = R^n e^{-n\epsilon}$$

where

$$R = \frac{\rho^{1/2}}{\left(\frac{b}{a}\left(a - \frac{\alpha}{2} + \frac{a}{b}r\right)\right)^{\frac{b}{a}(a-\frac{\alpha}{2}+\frac{a}{b}r)}}.$$

Now fix a $c > 0$ and let $n_t := [ct]$. Notice that $n_t \geq N_\epsilon$ for any $t > t_\epsilon := (N_\epsilon + 1)/c$.

Let $t > t_\epsilon$.

$$k^{n_t} \theta^{n_t/2} U_{n_t}(t) = k^{n_t} \theta^{n_t/2} t^{\frac{b}{a}(a-\frac{\alpha}{2}+\frac{a}{b}r)n_t} U_{n_t}(1) \geq \frac{k^{n_t} \theta^{n_t/2} t^{\frac{b}{a}(a-\frac{\alpha}{2}+\frac{a}{b}r)n_t}}{(n_t!)^{\frac{b}{a}(a-\frac{\alpha}{2}+\frac{a}{b}r)}} R^{n_t} e^{-n_t \epsilon}$$

Notice that $[ct]/t \rightarrow c$ as $t \rightarrow \infty$ which means that $n_t/t \rightarrow c$ as $t \rightarrow \infty$. With this we can say

$$\liminf_{t \rightarrow \infty} \frac{1}{t} \log \left(k^{n_t} \theta^{n_t/2} U_{n_t}(t) \right) = c \liminf_{t \rightarrow \infty} \frac{1}{n_t} \log \left(k^{n_t} \theta^{n_t/2} U_{n_t}(t) \right) =: I(n_t)$$

Now,

$$\begin{aligned} I(n_t) &\geq c \liminf_{t \rightarrow \infty} \frac{1}{n_t} \log \left((kR\sqrt{\theta})^{n_t} \frac{t^{\frac{b}{a}(a-\frac{\alpha}{2}+\frac{a}{b}r)n_t}}{(n_t!)^{\frac{b}{a}(a-\frac{\alpha}{2}+\frac{a}{b}r)}} \right) - c\epsilon \\ &= c \log(k\sqrt{\theta}R) + c \liminf_{t \rightarrow \infty} \frac{1}{n_t} \log \left[\left(\frac{t}{n_t} \right)^{\frac{b}{a}(a-\frac{\alpha}{2}+\frac{a}{b}r)n_t} \frac{n_t^{\frac{b}{a}(a-\frac{\alpha}{2}+\frac{a}{b}r)n_t}}{(n_t!)^{\frac{b}{a}(a-\frac{\alpha}{2}+\frac{a}{b}r)}} \right] - c\epsilon \\ &= c \log(k\sqrt{\theta}R) - c \frac{b}{a} \left(a - \frac{\alpha}{2} + \frac{a}{b}r \right) \log(c) + c \frac{b}{a} \left(a - \frac{\alpha}{2} + \frac{a}{b}r \right) \liminf_{t \rightarrow \infty} \frac{1}{n_t} \log \left(\frac{n_t^{n_t}}{(n_t)!} \right) - c\epsilon \\ &= c \log(k\sqrt{\theta}R) - c \frac{b}{a} \left(a - \frac{\alpha}{2} + \frac{a}{b}r \right) \log(c) + c \frac{b}{a} \left(a - \frac{\alpha}{2} + \frac{a}{b}r \right) - c\epsilon \end{aligned}$$

and letting ϵ tend to 0 we see that

$$I(n_t) \geq c \left[\log(k\sqrt{\theta}R) - \frac{b}{a} \left(a - \frac{\alpha}{2} + \frac{a}{b}r \right) \log(c) + \frac{b}{a} \left(a - \frac{\alpha}{2} + \frac{a}{b}r \right) \right] =: h(c).$$

In order to maximize $h(c)$, notice that

$$h'(c) = 0 \iff c^* = (k\sqrt{\theta}R)^{\frac{a}{b(a-\frac{\alpha}{2}+\frac{a}{b}r)}}.$$

After plugging c^* and replacing R we arrive at the following inequality

$$I(n_t) \geq (k\sqrt{\theta})^{\frac{a}{b(a-\frac{\alpha}{2}+\frac{a}{b}r)}} (\rho^{1/2})^{\frac{a}{b(a-\frac{\alpha}{2}+\frac{a}{b}r)}},$$

which proves (5.14) after some simplification. □

We are now ready to prove Theorem 5.1.

Proof of Theorem 5.1. By proposition 5.4, for and $p, q > 0$ with $p^{-1} + q^{-1} = 1$ we have that

$$\|u(t, 0)\|_p \geq \exp \left\{ -\frac{1}{2} t_p^\beta \|f\|_{\mathcal{H}}^2 \right\} \left| \sum_{n \geq 0} \theta^{n/2} W_n(t_p^\beta, \mathcal{F}f) \right|.$$

We now take the supremum over all $f \in \mathcal{H}_+$ with $\|f\|_{\mathcal{H}} = k > 0$. Recall that $W_n(t, \phi) = k^n W_n(t, \phi/k)$ for $\phi \in L^2(\mu)$ and non-negativity of $W_n(t, \cdot)$ on $\mathcal{H}_+$. Let $c > 0$.

$$\begin{aligned} \|u(t, 0)\|_p &\geq \exp \left\{ -\frac{1}{2} t_p^\beta k^2 \right\} \sup_{f \in \mathcal{H}_+ \mid \|f\|_{\mathcal{H}}=k} \sum_{n \geq 0} \theta^{n/2} W_n(t_p^\beta, \mathcal{F}f) \\ &= \exp \left\{ -\frac{1}{2} t_p^\beta k^2 \right\} \sup_{f \in \mathcal{H}_+ \mid \|f\|_{\mathcal{H}}=1} \sum_{n \geq 0} k^n \theta^{n/2} W_n(t_p^\beta, \mathcal{F}f) \\ &\geq \exp \left\{ -\frac{1}{2} t_p^\beta k^2 \right\} \sup_{f \in \mathcal{H}_+ \mid \|f\|_{\mathcal{H}}=1} k^{n_t} \theta^{n_t/2} U_{n_t}(t_p^\beta, f) \end{aligned}$$

where $n_t = \lceil c t_p^\beta \rceil$. Now by choosing c as in Lemma 5.7 we get that

$$\begin{aligned} \liminf_{n \rightarrow \infty} t_p^{-\beta} \log \|u(t, x)\|_p &\geq \liminf_{n \rightarrow \infty} \left(\frac{-\frac{1}{2} t_p^\beta k^2}{t_p^\beta} + \frac{1}{t_p^\beta} \log \left[\sup_{f \in \mathcal{H}_+ \mid \|f\|_{\mathcal{H}}=1} k^{n_t} \theta^{n_t/2} U_{n_t}(t_p^\beta, f) \right] \right) \\ &= -\frac{1}{2} k^2 + k^{\frac{2a}{2ab-\alpha b+2ar}} (\rho \theta)^{\frac{a}{2ab-\alpha b+2ar}} =: h(k). \end{aligned}$$

By maximizing h for $k > 0$, we see that h is maximized at the point

$$k^* = \left(\frac{2ab - \alpha b + 2ar}{2aB} \right)^{\frac{2ab - \alpha b + 2ar}{2a - 2[2ab - \alpha b + 2ar]}} \quad \text{with } B = (\theta \rho)^{\frac{a}{2ab - \alpha b + 2ar}}.$$

Inserting k^* into h gives us that

$$h(k^*) = B^\beta \left(\frac{2a}{2ab - \alpha b + 2ar} \right)^\beta \left(\frac{2ab - \alpha b + 2ar - a}{2a} \right)$$

and plugging in the value for B proves Theorem 5.1. $\square$

6 Appendix

S:Appendix

Proof of Lemma 2.4. Recall that we define $\varphi(x) = C_{\alpha,d} |x|^{-(d-\alpha)}$. We begin by applying the substitution $x' = (\frac{\nu}{2})^{1/a}$ to ρ :

$$\begin{aligned} \rho &= \sup_{\|f\|_2=1} \int_{\mathbb{R}^d} \left[\int_{\mathbb{R}^d} \frac{f(x+y)f(y)}{\sqrt{1 + \frac{\nu}{2}|x+y|^a} \sqrt{1 + \frac{\nu}{2}|y|^a}} dy \right]^2 \mu(dx) \\ &= \sup_{\|f\|_2=1} \int_{\mathbb{R}^d} \left[\int_{\mathbb{R}^d} \frac{f\left(\left(\frac{\nu}{2}\right)^{-1/a}(x+y)\right) f\left(\left(\frac{\nu}{2}\right)^{-1/a}y\right)}{\sqrt{1 + |x+y|^a} \sqrt{1 + |y|^a}} \left(\frac{\nu}{2}\right)^{-d/a} dy \right]^2 \left(\frac{\nu}{2}\right)^{(d-\alpha)/a} \varphi(x) \left(\frac{\nu}{2}\right)^{-d/a} dx \\ &= \left(\frac{\nu}{2}\right)^{-\alpha/a} \sup_{\|f\|_2=1} \int_{\mathbb{R}^d} \left[\int_{\mathbb{R}^d} \frac{f\left(\left(\frac{\nu}{2}\right)^{-1/a}(x+y)\right) f\left(\left(\frac{\nu}{2}\right)^{-1/a}y\right)}{\sqrt{1 + |x+y|^a} \sqrt{1 + |y|^a}} \left(\frac{\nu}{2}\right)^{-d/a} dy \right]^2 \varphi(x) dx \end{aligned}$$

Now define $f^*(x) = (\frac{\nu}{2})^{\frac{-d}{2a}} f\left((\frac{\nu}{2})^{-1/a} x\right)$. Then with this we see that

$$\rho = \left(\frac{\nu}{2}\right)^{-\alpha/a} \sup_{\|f\|_2=1} \int_{\mathbb{R}^d} \left[\int_{\mathbb{R}^d} \frac{f^*(x+y) f^*(y)}{\sqrt{1+|x+y|^a} \sqrt{1+|y|^a}} dy \right]^2 \varphi(x) dx$$

and note that

$$\int_{\mathbb{R}^d} f(x)^2 dx = \int_{\mathbb{R}^d} f^*(x)^2 dx$$

and thus

$$\left\{ f \in L^2(\mathbb{R}^d) : \|f\|_2 = 1 \right\} = \left\{ f \in L^2(\mathbb{R}^d) : \|f^*\|_2 = 1 \right\}$$

so we can say that

$$\begin{aligned} \rho &= \left(\frac{\nu}{2}\right)^{-\alpha/a} \sup_{\|f\|_2=1} \int_{\mathbb{R}^d} \left[\int_{\mathbb{R}^d} \frac{f^*(x+y) f^*(y)}{\sqrt{1+|x+y|^a} \sqrt{1+|y|^a}} dy \right]^2 \varphi(x) dx \\ &= \left(\frac{\nu}{2}\right)^{-\alpha/a} \sup_{\|f^*\|_2=1} \int_{\mathbb{R}^d} \left[\int_{\mathbb{R}^d} \frac{f^*(x+y) f^*(y)}{\sqrt{1+|x+y|^a} \sqrt{1+|y|^a}} dy \right]^2 \varphi(x) dx \\ &= \left(\frac{\nu}{2}\right)^{-\alpha/a} \rho_{2,a} \end{aligned}$$

and (2.15) is proven by applying (2.12).

To prove (2.16), we first notice that by applying the substitution $\xi'_{\sigma(j)} = (\frac{\nu}{2})^{1/a} \xi_{\sigma(j)}$ that

$$\begin{aligned} &\int_{(\mathbb{R}^d)^n} \left[\sum_{\sigma \in \Sigma_n} \prod_{k=1}^n \frac{1}{1 + \frac{\nu}{2} |\sum_{j=k}^n \xi_{\sigma(j)}|^a} \right]^2 \mu(d\vec{\xi}) \\ &= \left(\frac{\nu}{2}\right)^{-n\alpha/a} \int_{(\mathbb{R}^d)^n} \left[\sum_{\sigma \in \Sigma_n} \prod_{k=1}^n \frac{1}{1 + |\sum_{j=k}^n \xi_{\sigma(j)}|^a} \right]^2 \mu(d\vec{\xi}) \end{aligned}$$

and now we can directly calculate

$$\begin{aligned} &\lim_{n \rightarrow \infty} \frac{1}{n} \log \left[\frac{1}{(n!)^2} \int_{(\mathbb{R}^d)^n} \left(\sum_{\sigma \in \Sigma_n} \prod_{k=1}^n \frac{1}{1 + \frac{\nu}{2} |\sum_{j=k}^n \xi_{\sigma(j)}|^a} \right)^2 \mu(d\vec{\xi}) \right] \\ &= \lim_{n \rightarrow \infty} \frac{1}{n} \log \left[\frac{1}{(n!)^2} \left(\frac{\nu}{2}\right)^{-n\alpha/a} \int_{(\mathbb{R}^d)^n} \left[\sum_{\sigma \in \Sigma_n} \prod_{k=1}^n \frac{1}{1 + |\sum_{j=k}^n \xi_{\sigma(j)}|^a} \right]^2 \mu(d\vec{\xi}) \right] \\ &= \log \left[\left(\frac{\nu}{2}\right)^{-\alpha/a} \right] + \lim_{n \rightarrow \infty} \frac{1}{n} \log \left[\frac{1}{(n!)^2} \int_{(\mathbb{R}^d)^n} \left[\sum_{\sigma \in \Sigma_n} \prod_{k=1}^n \frac{1}{1 + |\sum_{j=k}^n \xi_{\sigma(j)}|^a} \right]^2 \mu(d\vec{\xi}) \right] \end{aligned}$$

and because of both Theorem 1.5 [3] and (2.15) above,

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \left(\frac{1}{(n!)^2} \int_{(\mathbb{R}^d)^n} \left(\sum_{\sigma \in \Sigma_n} \prod_{k=1}^n \frac{1}{1 + \frac{\nu}{2} |\sum_{j=k}^n \xi_{\sigma(j)}|^a} \right)^2 \mu(d\vec{\xi}) \right)$$

$$= \log \left(\left(\frac{\nu}{2} \right)^{-\alpha/a} \right) + \log (\rho_{2,a}) = \log (\rho_{\nu,a}) .$$

This completes the proof of Lemma 2.4. □

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